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Nehari Manifold for Fractional Kirchhoff Systems with Critical Nonlinearity

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Abstract. In this paper, we show the existence and multiplicity of positive solutions of the fractional Kirchhoff system

$$\begin{cases} \mathcal{L}_M(u) = \lambda f(x)|u|^{q-2}u + \frac{2\alpha}{\alpha+\beta}\,|u|^{\alpha-2}\,u|v|^{\beta} & \text{in } \Omega, \\ \mathcal{L}_M(v) = \mu g(x)|v|^{q-2}v + \frac{2\beta}{\alpha+\beta}\,|u|^{\alpha}\,|v|^{\beta-2}v & \text{in } \Omega, \\ \quad u = v = 0 & \text{on } \partial\Omega, \end{cases}$$

where $\mathcal{L}_M(u) = M\left(\int_\Omega |(-\Delta)^{\frac{s}{2}}u|^2dx\right)(-\Delta)^s u$ is a double non-local operator due to a Kirchhoff term $M(t) = a + bt$ with $a, b > 0$ and the fractional Laplacian $(-\Delta)^s$, $s \in (0, 1)$. We consider that Ω is an open and bounded domain in $\mathbb{R}^N$, $2s < N \leq 4s$ with smooth boundary, f, g are sign-changing continuous functions, $\lambda, \mu > 0$ are real parameters, $1 < q < 2$, $\alpha, \beta \geq 2$ and $\alpha + \beta = 2^*_s = 2N/(N-2s)$ is a fractional critical exponent. Using the idea of Nehari manifold technique and a compactness result based on the classical idea of the Brezis–Lieb lemma, we prove the existence of at least two positive solutions for (λ, μ) lying in a suitable subset of $\mathbb{R}^2_+$.

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Keywords. Nehari manifold, fractional Kirchhoff system, critical exponent, multiplicity.

1. Introduction

In this paper, we show the existence and multiplicity of positive solutions for the fractional Kirchhoff system

$$(P_{\lambda,\mu}) \begin{cases} \mathcal{L}_M(u) = \lambda f(x)|u|^{q-2}u + \frac{2\alpha}{\alpha+\beta}\,|u|^{\alpha-2}\,u|v|^{\beta} & \text{in } \Omega, \\ \mathcal{L}_M(v) = \mu g(x)|v|^{q-2}v + \frac{2\beta}{\alpha+\beta}\,|u|^{\alpha}\,|v|^{\beta-2}v & \text{in } \Omega, \\ \quad u = v = 0 & \text{on } \partial\Omega, \end{cases}$$

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where $\mathcal{L}_M(u) = M\left(\int_\Omega |(-\Delta)^{\frac{s}{2}}u|^2dx\right)(-\Delta)^s u$ is a double non-local operator emerged from the fusion of a Kirchhoff term $M(t) = a + bt$ with $a, b > 0$ and spectral fractional Laplacian $(-\Delta)^s$, $s \in (0,1)$ (see [13] for modeling of a vibrating string which gives rise to this kind of operators). We consider that Ω is an open and bounded domain in $\mathbb{R}^N$, $2s < N \leq 4s$ with smooth boundary, $\lambda, \mu > 0$ are real parameters, $1 < q < 2$, $\alpha, \beta \geq 2$ and $\alpha + \beta = 2_s^* = 2N/(N-2s)$ is a fractional critical exponent.

Define $f^+(x) = \max\{f(x), 0\}$ and $g^+(x) = \max\{g(x), 0\}$. Now for the ease of reference, we impose the following condition on the continuous weights f, g:

(fg) $\quad f, g \in L^{\gamma_*}(\Omega)$, where $\gamma_* > \gamma = 2_s^*/(2_s^* - q)$ and $f^+, g^+ \not\equiv 0$.

In this work, we prove the multiplicity of positive solutions for the system $(P_{\lambda,\mu})$ using the popular technique of constrained minimization on an Nehari manifold. We show the multiplicity result by extracting Palais–Smale sequences in the non-empty decompositions of the Nehari manifold for (λ, μ) lying in some suitable subset of $\mathbb{R}^2_+$. In the presence of the critical nonlinearity (the lack of compactness of the critical Sobolev embedding), the Kirchhoff term may not allow the weak limit of the Palais–Smale sequence to be a weak solution of the problem. This makes this class of problems interesting to study.

To overcome the nonlocal behaviour caused by the nonlocal fractional operator, we have adopted the celebrated idea of the Caffarelli–Silvestre extension, see [7] but in the process we lack the explicit form of extremal functions for critical fractional Sobolev embeddings. We have used some suitable asymptotic estimates of these extremals, obtained in [3, 22]. Apart from these challenges, due to a critical growth, we lack the compactness which is needed in case of $M \not\equiv 1$ to get the pre-compactness of Palais–Smale sequences. Based on the classical idea of the Brezis–Lieb lemma, we have derived a compactness result (see Proposition 4.1 in Section 4) to prove the main result of this paper (see Theorem 1.1 below).

For $u = v$, $\alpha = \beta$, $\lambda = \mu$ and $f = g$, the problem $(P_{\lambda,\mu})$ reduces into the scalar case. We cite [12], for related results in case of scalar equation, where authors have applied a classical idea of a concentration-compactness lemma due to P.L. Lions. In the local setting and in the absence of the Kirchhoff term, the systems of linear and quasilinear elliptic equations have been studied in [1, 2, 4, 23] and references therein.

In the case $M \equiv 1$ the fractional systems have been studied in [9, 17]. Precisely, for $f = g = 1$, in [17], the authors have studied the following system with critical growth:

$$
\begin{aligned}
(-\Delta)^s u &= \lambda |u|^{q-2}u + \frac{2\alpha}{\alpha+\beta}|u|^{\alpha-2}u|v|^\beta \text{ in } \Omega,\\
(-\Delta)^s v &= \mu |v|^{q-2}v + \frac{2\beta}{\alpha+\beta}|u|^{\alpha}|v|^{\beta-2}v \text{ in } \Omega,\\
u &= v = 0 \text{ in } \mathbb{R}^N \setminus \Omega,
\end{aligned}
$$

where $\alpha, \beta > 1$ and $\alpha + \beta = 2_s^*$, $\Omega \subset \mathbb{R}^N$ is an open bounded domain with smooth boundary, $N > 2s, 0 < s < 1$, $1 < q < 2$. They proved the multiplicity of solutions using the idea of a Nehari manifold and harmonic extension for suitable choice of $\lambda, \mu > 0$. The extension of these results for quasilinear problems with critical growth can be seen in [10].

However, in the case of Kirchhoff fractional systems, in [9, 24], the authors have obtained the multiplicity of solutions for a p-fractional Kirchhoff system in the subcritical case for any $p \geq 2$. We emphasize that in these papers the non-local operator is taken in the integrodifferential form which is different from our spectral definition (see [21] for discussion) and require different boundary conditions. In addition the Kirchhoff term involved both norms of u and v in [24]. Existence results in the critical case are discussed in [14, Theorem 1.2] and in [19, Theorem 1.3]. To the best of our knowledge, there are no results for the multiplicity of fractional Kirchhoff systems with critical nonlinearity.

Define the set

$$\mathcal{M}_\Gamma = \left\{ (\lambda, \mu) \in \mathbb{R}_+^2 : \left((\lambda \|f\|_\gamma)^{\frac{2}{2-q}} + (\mu \|g\|_\gamma)^{\frac{2}{2-q}} \right)^{\frac{2-q}{2}} < \Gamma \right\}, \tag{1.1}$$

where $\|\cdot\|_\gamma$ is the notation used for the standard norm in $L^\gamma(\Omega)$. In the presence of critical nonlinearity, we prove the following theorem.

Theorem 1.1. *Assume $a > 0$ and $b > 0$. Then:*

(i) *There exists a $\Gamma_0 > 0$ such that problem $(P_{\lambda,\mu})$ has at least one positive solution for $(\lambda, \mu) \in \mathcal{M}_{\Gamma_0}$ with negative energy.*

(ii) *Assume in addition that $\{x \in \Omega \,|\, f(x) > 0$ and $g(x) > 0\}$ has a non-zero measure. Then, for $b \in (0, b_0)$ for some $b_0 > 0$, there exists $0 < \Gamma_{00} = \Gamma_{00}(b) < \Gamma_0$ such that for $(\lambda, \mu) \in \mathcal{M}_{\Gamma_{00}}$ the problem $(P_{\lambda,\mu})$ has at least two positive solutions and among them one has positive energy.*

Theorem 1.1 complements the result of [9] for the critical case and [24] for the non-degenerate Kirchhoff case with critical nonlinearity in low dimensions. By overcoming the lack of compactness, we have obtained two nontrivial positive solutions by avoiding the possibility of semi-trivial solutions.

Remark 1.2. We remark that the restriction on b in Theorem 1.1 (ii) is required while showing that the Palais–Smale level in a suitable Nehari subset lies within the compactness level (see Lemma 6.1).

Remark 1.3. We remark that the multiplicity result of Theorem 1.1 (ii) holds true for all $b > 0$ when the pair (N, s) takes the following values.

$$\left(1, \frac{1}{4}\right), \left(2, \frac{1}{2}\right), \left(3, \frac{3}{4}\right), \left(1, \frac{1}{3}\right), \left(2, \frac{2}{3}\right).$$

Observe that in these cases either $2_s^* = 4$ or $2_s^* = 6$ which allows us to find an explicit positive root of an algebraic polynomial of degree 2_s^* involved in the estimation of the energy level in the Nehari manifold (see Lemma 6.1). Since 2_s^* is not an integer, in general, it is not always possible to find a zero of an algebraic expression.

Remark 1.4. The restriction $2s < N \leq 4s$ (or $2_s^* \geq 4$) can be removed if we consider a general Kirchhoff model as

$$M(t) = a + bt^{\theta-1}, \quad \text{with } a > 0 \text{ and } b > 0, \text{ for any } t \geq 0, \text{ and } \theta \in [1, 2_s^*/2),$$

which includes the standard Kirchhoff case (for $\theta = 2$) as well. For the ease of presentation of the technique we have considered only the standard Kirchhoff case.

The paper is organized as follows. In Section 2, preliminaries and variational settings are defined. In Section 3, we introduce the Nehari manifold and analysis of fibering maps. In Section 4, we study the compactness of Palais–Smale sequences extracted from Nehari decompositions. In Section 5, the existence of the first solution is proved. Finally, in Section 6 we have shown the existence of the second solution and concluded the proof of Theorem 1.1. A proof of an elementary inequality is added in the Appendix for the sake of completeness.

2. Variational Formulation

The fractional powers of the Laplacian $(-\Delta)^s$ in a bounded domain Ω with zero Dirichlet boundary data are defined through the spectral decomposition using the powers of the eigenvalues of the Laplacian operator. Let (φ_j, ρ_j) be the eigenfunctions and eigenvectors of $(-\Delta)$ in Ω with zero Dirichlet boundary data. Then (φ_j, ρ_j^s) are the eigenfunctions and eigenvectors of $(-\Delta)^s$ with Dirichlet boundary conditions. In fact, the fractional Laplacian $(-\Delta)^s$ is well defined in the space of functions

$$H_0^s(\Omega) = \left\{ u = \sum c_j \varphi_j \in L^2(\Omega) : \ \|u\|_{H_0^s(\Omega)} = \left(\sum c_j^2 \rho_j^s \right)^{1/2} < \infty \right\}$$

and, as a consequence,

$$(-\Delta)^s u = \sum c_j \rho_j^s \varphi_j \, .$$

Note that $\|u\|_{H_0^s(\Omega)} = \|(-\Delta)^{s/2} u\|_2$. The dual space $H^{-s}(\Omega)$ and the inverse operator $(-\Delta)^{-s}$ are defined in the standard way. The energy functional associated to $(P_{\lambda,\mu})$ is given by

$$\begin{aligned} \mathcal{J}_{\lambda,\mu}(u,v) = & \frac{1}{2}\widehat{M}\left(\int_\Omega |(-\Delta)^{\frac{s}{2}} u|^2 dx \right) + \frac{1}{2}\widehat{M}\left(\int_\Omega |(-\Delta)^{\frac{s}{2}} v|^2 dx \right) \\ & - \frac{\lambda}{q} \int_\Omega f(x)|u|^q dx - \frac{\mu}{q} \int_\Omega g(x)|v|^q dx - \frac{2}{\alpha+\beta} \int_\Omega |u|^\alpha |v|^\beta dx, \end{aligned}$$

where $\widehat{M}(t) = \int_0^t M(s) ds$ is the primitive of M. Under **(fg)**, it is easy to see that the functional $\mathcal{J}_{\lambda,\mu}$ is well defined and continuously differentiable on $H_0^s(\Omega) \times H_0^s(\Omega)$.

Definition 2.1. A function $(u,v) \in H_0^s(\Omega) \times H_0^s(\Omega)$ is called a weak solution of $(P_{\lambda,\mu})$ if for all $(\varphi,\psi) \in H_0^s(\Omega) \times H_0^s(\Omega)$

$$M\left(\int_\Omega |(-\Delta)^{\frac{s}{2}}u|^2dx\right)\int_\Omega (-\Delta)^{\frac{s}{2}}u(-\Delta)^{\frac{s}{2}}\varphi dx$$
$$+ M\left(\int_\Omega |(-\Delta)^{\frac{s}{2}}v|^2dx\right)\int_\Omega (-\Delta)^{\frac{s}{2}}v(-\Delta)^{\frac{s}{2}}\psi dx$$
$$= \lambda\int_\Omega f(x)|u|^{q-2}u\varphi\, dx + \mu\int_\Omega g(x)|v|^{q-2}v\psi\, dx + \frac{2\alpha}{\alpha+\beta}\int_\Omega |u|^{\alpha-2}\, u|v|^\beta\varphi dx$$
$$|quad + \frac{2\beta}{\alpha+\beta}\int_\Omega |u|^\alpha\, |v|^{\beta-2}v\psi dx.$$

Now we discuss a harmonic extension technique developed by Caffarelli and Silvestre [7] to treat the non-local problems involving fractional Laplacians. In this technique, we study an extension problem corresponding to a non-local problem so that we can investigate the non-local problem via classical variational methods. To begin with, we first define the harmonic extension of $u \in H_0^s(\Omega)$ as follows.

Definition 2.2. For $u \in H_0^s(\Omega)$, the harmonic extension $E_s(u) := w$ is the solution of the problem

$$\begin{cases} -\text{div}(y^{1-2s}\nabla w) = 0 & \text{in} \quad \Omega\times(0,\infty) =: \mathcal{C}, \\ w = 0 & \text{on} \quad \partial\Omega\times(0,\infty) =: \partial_L, \\ w = u & \text{on} \quad \Omega\times\{0\}. \end{cases}$$

Moreover, the extension function is related to the fractional Laplacian by

$$(-\Delta)^s u(x) = -\kappa_s \lim_{y\to 0^+} y^{1-2s}\frac{\partial w}{\partial y}(x,y),$$

where $\kappa_s = 2^{2s-1}\frac{\Gamma(s)}{\Gamma(1-s)}$.

The solution space for the extension problem is the Hilbert space

$$E_0^1(\mathcal{C}) = \left\{w \in L^2(\mathcal{C}):\ w = 0 \text{ on } \partial_L,\ \|w\| < \infty\right\}$$

where the norm

$$\|w\| := \left(\kappa_s\int_{\mathcal{C}} y^{1-2s}|\nabla w|^2dxdy\right)^{1/2}$$

is induced on $E_0^1(\mathcal{C})$ through the inner product

$$\langle w, z\rangle = \kappa_s\int_{\mathcal{C}} y^{1-2s}\nabla w.\nabla z dx\, dy.$$

We observe that the extension operator is an isometry between $H_0^s(\Omega)$ and $E_0^1(\mathcal{C})$. That is, for all $u \in H_0^s(\Omega)$, we have

$$\|E_s(u)\| = \|u\|_{H_0^s(\Omega)}. \tag{2.1}$$

This isometry in (2.1) is the key to study the Kirchhoff-type problems in the harmonic extension setup. Denote $E_s(u)=w$ and $E_s(v)=z$ in the sense of Definition 2.2. Then, the problem $(P_{\lambda,\mu})$ is equivalent to studying the extension problem

$$(S_{\lambda,\mu})\begin{cases} -\mathrm{div}(y^{1-2s}\nabla w)=0,\ -\mathrm{div}(y^{1-2s}\nabla z)=0, & \text{in } \mathcal{C},\\ w=z=0, & \text{on } \partial_L,\\ M(\|w\|^2)\frac{\partial w}{\partial \nu}=\lambda f|w|^{q-2}w+\frac{2\alpha}{\alpha+\beta}\,|w|^{\alpha-2}\,w|z|^{\beta}, & \text{on } \Omega\times\{0\},\\ M(\|z\|^2)\frac{\partial z}{\partial \nu}=\mu g|z|^{q-2}z+\frac{2\beta}{\alpha+\beta}\,|w|^{\alpha}\,|z|^{\beta-2}z, & \text{on } \Omega\times\{0\},\end{cases}$$

where $\dfrac{\partial w}{\partial \nu}=-\kappa_s\displaystyle\lim_{y\to 0^+}y^{1-2s}\frac{\partial w}{\partial y}(x,y)$ and $\frac{\partial z}{\partial \nu}=-\kappa_s\displaystyle\lim_{y\to 0^+}y^{1-2s}\frac{\partial z}{\partial y}(x,y)$.

The natural space to look for the solution of $(S_{\lambda,\mu})$ is the product space $\mathcal{H}(\mathcal{C}):=E_0^1(\mathcal{C})\times E_0^1(\mathcal{C})$ endowed with the product norm

$$\|(w,z)\|^2=\kappa_s\left(\int_{\mathcal{C}}y^{1-2s}|\nabla w|^2dxdy+\int_{\mathcal{C}}y^{1-2s}|\nabla z|^2dxdy\right).$$

The variational functional $\mathcal{I}_{\lambda,\mu}:\mathcal{H}(\mathcal{C})\to\mathbb{R}$ associated to $(S_{\lambda,\mu})$ is defined as

$$\begin{aligned}\mathcal{I}_{\lambda,\mu}(w,z)&=\frac{1}{2}\widehat{M}(\|w\|^2)+\frac{1}{2}\widehat{M}(\|z\|^2)-\frac{\lambda}{q}\int_\Omega f(x)|w(x,0)|^q dx\\ &|quad-\frac{\mu}{q}\int_\Omega g(x)|z(x,0)|^q dx-\frac{2}{\alpha+\beta}\int_\Omega|w(x,0)|^\alpha|z(x,0)|^\beta dx.\end{aligned}$$

Without great efforts it can be shown that $\mathcal{I}_{\lambda,\mu}$ is well defined and C^1. Now we give the definition of a weak solution of the extension problem $(S_{\lambda,\mu})$.

Definition 2.3. A function $(w,z)\in\mathcal{H}(\mathcal{C})$ is called a weak solution of $(S_{\lambda,\mu})$ if for all $(\varphi,\psi)\in\mathcal{H}(\mathcal{C})$

$$\begin{aligned}&M\left(\|w\|^2\right)\kappa_s\int_{\mathcal{C}}y^{1-2s}\nabla w.\nabla\varphi dxdy+M\left(\|z\|^2\right)\kappa_s\int_{\mathcal{C}}y^{1-2s}\nabla z.\nabla\psi dxdy\\ &-\lambda\int_\Omega f(x)|w(x,0)|^{q-2}w(x,0)\varphi(x,0)\,dx-\mu\int_\Omega g(x)|z(x,0)|^{q-2}z(x,0)\psi(x,0)\,dx\\ &-\frac{2\alpha}{\alpha+\beta}\int_\Omega|w(x,0)|^{\alpha-2}\,w(x,0)|z(x,0)|^\beta\varphi(x,0)dx\\ &-\frac{2\beta}{\alpha+\beta}\int_\Omega|w(x,0)|^\alpha\,|z(x,0)|^{\beta-2}z(x,0)\psi(x,0)dx=0.\end{aligned}$$

It is clear that the critical points of $\mathcal{I}_{\lambda,\mu}$ in $\mathcal{H}(\mathcal{C})$ correspond to the critical points of $\mathcal{J}_{\lambda,\mu}$ in $H_0^s(\Omega)\times H_0^s(\Omega)$. Thus if (w,z) solves $(S_{\lambda,\mu})$, then (u,v), where $u=\text{trace }(w)=w(x,0)$, $v=\text{trace }(z)=z(x,0)$ is a solution of $(P_{\lambda,\mu})$ and vice versa.

Now, we state the following trace inequality which will be used in the subsequent lemmas. From (2.1) and Sobolev embeddings (see also [5, Theorems 2.1 and 4.4]) we have:

Lemma 2.4. *Let $2 \leq r \leq 2_s^*$, then there exists $C_r > 0$ such that for all $w \in E_0^1(\mathcal{C})$,*

$$\int_{\mathcal{C}} y^{1-2s} |\nabla w|^2 dxdy \geq C_r \left(\int_{\Omega} |w(x,0)|^r dx \right)^{\frac{2}{r}}. \tag{2.2}$$

Moreover for $r = 2_s^*$, the best constant in (2.2) will be denoted by $S(s,N)$, that is,

$$S(s,N) := \inf_{w \in E_0^1(\mathcal{C}) \setminus \{0\}} \frac{\displaystyle\int_{\mathcal{C}} y^{1-2s} |\nabla w|^2 dxdy}{\left(\displaystyle\int_{\Omega} |w(x,0)|^{2_s^*} dx \right)^{\frac{2}{2_s^*}}}, \tag{2.3}$$

and it is indeed achieved in the case when $\Omega = \mathbb{R}^N$ and $w = E_s(u)$, where

$$u(x) = u_\varepsilon(x) = \frac{\varepsilon^{(N-2s)/2}}{(|x|^2 + \varepsilon^2)^{(N-2s)/2}} \tag{2.4}$$

with $\varepsilon > 0$ arbitrary.

We conclude this section by introducing the following minimization problem:

$$S(s,\alpha,\beta) := \inf_{(w,z) \in \mathcal{H}(\mathcal{C}) \setminus \{0\}} \frac{\displaystyle\int_{\mathcal{C}} y^{1-2s} (|\nabla w|^2 + |\nabla z|^2) dxdy}{\left(\displaystyle\int_{\Omega} |w(x,0)|^{\alpha} |z(x,0)|^{\beta} dx \right)^{\frac{2}{2_s^*}}}. \tag{2.5}$$

In light of the inequality $|w|^\alpha |z|^\beta \leq |w|^{\alpha+\beta} + |z|^{\alpha+\beta}$ and Lemma 2.4, the best Sobolev constant in (2.5) is well defined. Using the ideas from [2], the authors of [17] established the following relationship between $S(s,N)$ and $S(s,\alpha,\beta)$.

Lemma 2.5. *For the constants $S(s,N)$ and $S(s,\alpha,\beta)$ introduced in (2.3) and (2.5), respectively, it holds*

$$S(s,\alpha,\beta) = \left[\left(\frac{\alpha}{\beta} \right)^{\frac{\beta}{\alpha+\beta}} + \left(\frac{\beta}{\alpha} \right)^{\frac{\alpha}{\alpha+\beta}} \right] S(s,N).$$

In particular, the constant $S(s,\alpha,\beta)$ is achieved for $\Omega = \mathbb{R}^N$.

3. Nehari manifold for $(S_{\lambda,\mu})$

In this section we study the nature of a Nehari manifold associated with $(S_{\lambda,\mu})$. Observe that, in the case of $\alpha + \beta \geq 4$, the functional $\mathcal{I}_{\lambda,\mu}$ is not bounded below on $\mathcal{H}(\mathcal{C})$ which sparks the idea of constrained minimization on a non-empty subset of $\mathcal{H}(\mathcal{C})$, popularly known as Nehari manifold. We will show that the functional $\mathcal{I}_{\lambda,\mu}$ is bounded below on $\mathcal{N}_{\lambda,\mu}$ (see Lemma 3.3 below), and on minimizing $\mathcal{I}_{\lambda,\mu}$ on these subsets, we get the solutions of $(S_{\lambda,\mu})$ by showing that the minimizers on $\mathcal{N}_{\lambda,\mu}$ are the critical points of $\mathcal{I}_{\lambda,\mu}$ on $\mathcal{H}(\mathcal{C})$ (see Lemma 3.1 below).

We define the Nehari set $\mathcal{N}_{\lambda,\mu}$ as

$$\mathcal{N}_{\lambda,\mu} = \{(w,z) \in \mathcal{H}(\mathcal{C}) \setminus \{0\} : \left\langle \mathcal{I}'_{\lambda,\mu}(w,z), (w,z) \right\rangle_* = 0\},$$

where $\langle , \rangle_*$ is the duality between $\mathcal{H}(\mathcal{C})$ and its dual space. Thus $(w,z) \in \mathcal{N}_{\lambda,\mu}$ if and only if

$$\begin{aligned} M(\|w\|^2)\|w\|^2 + M(\|z\|^2)\|z\|^2 = \lambda \int_\Omega f(x)|w(x,0)|^q dx \\ + \mu \int_\Omega g(x)|z(x,0)|^q dx + 2\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx. \end{aligned} \tag{3.1}$$

Now for a fixed $(w,z) \in \mathcal{H}(\mathcal{C})$, we define $\phi_{(w,z)} : \mathbb{R}^+ \to \mathbb{R}$, much known as fiber maps, as $\phi_{(w,z)}(t) = \mathcal{I}_{\lambda,\mu}(tw,tz)$. Thus

$$\begin{aligned} \phi_{(w,z)}(t) = \frac{1}{2}\widehat{M}(t^2\|w\|^2) + \frac{1}{2}\widehat{M}(t^2\|z\|^2) - \frac{t^q}{q}\lambda \int_\Omega f(x)|w(x,0)|^q dx \\ - \frac{t^q}{q}\mu \int_\Omega g(x)|z(x,0)|^q dx - \frac{2t^{2_s^*}}{2_s^*}\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx. \end{aligned}$$

Now, for a fixed $(w,z) \in \mathcal{H}(\mathcal{C})$, taking the derivative with respect to the variable t and putting $t = 1$, we get

$$\begin{aligned} \phi'_{(w,z)}(1) = M(\|w\|^2)\|w\|^2 + M(\|z\|^2)\|z\|^2 - \lambda \int_\Omega f(x)|w(x,0)|^q dx \\ - \mu \int_\Omega g(x)|z(x,0)|^q dx - 2\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx, \end{aligned} \tag{3.2}$$

$$\begin{aligned} \phi''_{(w,z)}(1) = a(\|w\|^2 + \|z\|^2) + 3b(\|w\|^4 + \|z\|^4) - (q-1)\lambda \int_\Omega f(x)|w(x,0)|^q dx \\ - (q-1)\mu \int_\Omega g(x)|z(x,0)|^q dx - 2(2_s^* - 1)\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx. \end{aligned} \tag{3.3}$$

From equation (3.2), $(w,z) \in \mathcal{N}_{\lambda,\mu}$ if and only if $t = 1$ is a critical point of $\phi_{(w,z)(t)}$, that is, $\phi'_{(w,z)}(1) = 0$. Thus it is natural to split $\mathcal{N}_{\lambda,\mu}$ into three parts based on the classification of $t = 1$ as local minima, local maxima or points of inflection of the fibering maps, $\phi_{(w,z)}(t)$. For this, we set

$$\begin{aligned} \mathcal{N}^{\pm}_{\lambda,\mu} &:= \left\{ (w,z) \in \mathcal{N}_{\lambda,\mu} : \phi''_{(w,z)}(1) \gtrless 0 \right\}, \\ \mathcal{N}^{0}_{\lambda,\mu} &:= \left\{ (w,z) \in \mathcal{N}_{\lambda,\mu} : \phi''_{(w,z)}(1) = 0 \right\}. \end{aligned}$$

The following lemma shows that minimizers for $\mathcal{I}_{\lambda,\mu}$ restricted to $\mathcal{N}_{\lambda,\mu}$ are critical points for $\mathcal{I}_{\lambda,\mu}$. Thus the Nehari manifold is a natural constraint.

Lemma 3.1. *If (w,z) is a minimizer of $\mathcal{I}_{\lambda,\mu}$ on $\mathcal{N}_{\lambda,\mu}$ such that $(w,z) \notin \mathcal{N}^0_{\lambda,\mu}$, then (w,z) is a critical point for $\mathcal{I}_{\lambda,\mu}$.*

Proof. Let (w,z) be a local minimizer for $\mathcal{I}_{\lambda,\mu}$ in any of the subsets of $\mathcal{N}_{\lambda,\mu}$. Then, in any case (w,z) is a minimizer for $\mathcal{I}_{\lambda,\mu}$ under the constraint

$$\mathcal{F}_{\lambda,\mu}(w,z) := \left\langle \mathcal{I}'_{\lambda,\mu}(w,z), (w,z) \right\rangle_* = 0.$$

Since $(w,z) \notin \mathcal{N}^0_{\lambda,\mu}$, the constraint is non-degenerate in (w,z), then by the Lagrange multipliers rule, there exists $\eta \in \mathbb{R}$ such that $\mathcal{I}'_{\lambda,\mu}(w,z) = \eta \mathcal{F}'_{\lambda,\mu}(w,z)$. Thus

$\langle \mathcal{I}'_{\lambda,\mu}(w,z),(w,z)\rangle = \eta \left\langle \mathcal{F}'_{\lambda,\mu}(w,z),(w,z)\right\rangle_* = \eta\phi''_{(w,z)}(1)=0$, but $(w,z) \notin \mathcal{N}^0_{\lambda,\mu}$ and so $\phi''_{(w,z)}(1) \neq 0$. Hence $\eta = 0$ which completes the proof of the lemma. $\square$

Denote

$$\Gamma_1 := \left(\frac{a(2_s^* - 2)(\kappa_s S(s,N))^{\frac{q}{2}}}{2_s^* - q}\right)\left(\frac{a(2-q)(\kappa_s S(s,\alpha,\beta))^{\frac{2_s^*}{2}}}{2(2_s^* - q)}\right)^{\frac{2-q}{2_s^*-2}}. \tag{3.4}$$

The next lemma helps us to show that $\mathcal{N}_{\lambda,\mu}$ is a manifold for suitable choice of (λ,μ).

Lemma 3.2. *Let* $(\lambda,\mu) \in \mathcal{M}_{\Gamma_1}$, *where* $\mathcal{M}_\Gamma$ *is defined in* (1.1) *and* Γ_1 *as in* (3.4), *then* $\mathcal{N}^0_{\lambda,\mu} = \emptyset$.

Proof. We prove this lemma by contradiction. Assume on the contrary that $(w,z) \in \mathcal{N}^0_{\lambda,\mu}$. Then we have two cases.

Case 1: $(w,z) \in \mathcal{N}^0_{\lambda,\mu}$ and $\lambda \int_\Omega f(x)|w(x,0)|^q dx + \mu \int_\Omega g(x)|z(x,0)|^q dx = 0$.

From (3.2) and (3.3) we get,

$$\begin{aligned} a(\|w\|^2 + \|z\|^2) + 3b(\|w\|^4 + \|z\|^4) - 2(2_s^* - 1)\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx \\ = (2 - 2_s^*)a(\|w\|^2 + \|z\|^2) + (4 - 2_s^*)b(\|w\|^4 + \|z\|^4) < 0 \end{aligned}$$

which is a contradiction since $2_s^* \geq 4$.

Case 2: $(w,z) \in \mathcal{N}^0_{\lambda,\mu}$ and $\lambda \displaystyle\int_\Omega f(x)|w(x,0)|^q dx + \mu \int_\Omega g(x)|z(x,0)|^q dx \neq 0$.

Again from (3.2) and (3.3), we have

$$\begin{aligned} (2-q)a(\|w\|^2 + \|z\|^2) + (4-q)b(\|w\|^4 + \|z\|^4) \\ = 2(2_s^* - q)\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx \end{aligned} \tag{3.5}$$

and

$$\begin{aligned} (2_s^* - 2)a(\|w\|^2 + \|z\|^2) + (2_s^* - 4)b(\|w\|^4 + \|z\|^4) \\ = (2_s^* - q)\left(\lambda \int_\Omega f(x)|w(x,0)|^q dx + \mu \int_\Omega g(x)|z(x,0)|^q dx\right). \end{aligned} \tag{3.6}$$

Now define $\mathcal{T}_{\lambda,\mu} : \mathcal{N}_{\lambda,\mu} \to \mathbb{R}$ as

$$\begin{aligned} \mathcal{T}_{\lambda,\mu}(w,z) = \frac{(2_s^* - 2)a(\|w\|^2 + \|z\|^2) + (2_s^* - 4)b(\|w\|^4 + \|z\|^4)}{(2_s^* - q)} \\ - \lambda \int_\Omega f(x)|w(x,0)|^q dx - \mu \int_\Omega g(x)|z(x,0)|^q dx, \end{aligned} \tag{3.7}$$

then from (3.6) $\mathcal{T}_{\lambda,\mu}(w,z)=0$ for all $(w,z)\in\mathcal{N}^0_{\lambda,\mu}$. Also using (2.3), we have

$$\begin{aligned}\mathcal{T}_{\lambda,\mu}(w,z) &\geq \left(\frac{2^*_s-2}{2^*_s-q}\right)a(\|w\|^2+\|z\|^2)\\ &\quad -\lambda\int_\Omega f(x)|w(x,0)|^q dx-\mu\int_\Omega g(x)|z(x,0)|^q dx\\ &\geq \left(\frac{2^*_s-2}{2^*_s-q}\right)a(\|w\|^2+\|z\|^2)\\ &\quad -(\kappa_s S(s,N))^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}}+(\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}(\|w\|^2+\|z\|^2)^{\frac{q}{2}},\\ &\geq (\|w\|^2+\|z\|^2)^{\frac{q}{2}}\left(\left(\frac{2^*_s-2}{2^*_s-q}\right)a(\|w\|^2+\|z\|^2)^{\frac{2-q}{2}}\right.\\ &\quad \left.-(\kappa_s S(s,N)^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}}+(\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}\right).\end{aligned}$$

Now from (3.5) and (2.5), we get

$$\|w\|^2+\|z\|^2\geq\left(\frac{a(2-q)(\kappa_s S(s,\alpha,\beta))^{\frac{2^*_s}{2}}}{2(2^*_s-q)}\right)^{\frac{2}{2^*_s-2}}. \tag{3.8}$$

Using (3.8), we have

$$\begin{aligned}\mathcal{T}_{\lambda,\mu}(w,z)\geq (\|w\|^2+\|z\|^2)^{\frac{q}{2}}&\left(\left(\frac{a(2^*_s-2)}{2^*_s-q}\right)\left(\frac{a(2-q)(\kappa_s S(s,\alpha,\beta))^{\frac{2^*_s}{2}}}{2(2^*_s-q)}\right)^{\frac{2-q}{2^*_s-2}}\right.\\ &\left.-(\kappa_s S(s,N))^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}}+(\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}\right).\end{aligned}$$

This implies that for $(\lambda,\mu)\in\mathcal{M}_{\Gamma_1}$, $\mathcal{T}_{\lambda,\mu}(w,z)>0$, for all $(w,z)\in\mathcal{N}^0_{\lambda,\mu}$, which is a contradiction. □

Let us use the notations $\Theta_{\lambda,\mu}:=\inf\{\mathcal{I}_{\lambda,\mu}(w,z)|(w,z)\in\mathcal{N}_{\lambda,\mu}\}$ and $\Theta^{\pm}_{\lambda,\mu}:=\inf\{\mathcal{I}_{\lambda,\mu}(w,z)|(w,z)\in\mathcal{N}^{\pm}_{\lambda,\mu}\}$. The next lemma justifies that the above minimization problems on $\mathcal{N}_{\lambda,\mu}$ are well defined.

Lemma 3.3. *$\mathcal{I}_{\lambda,\mu}$ is coercive and bounded below on $\mathcal{N}_{\lambda,\mu}$.*

Proof. For $(w,z)\in\mathcal{N}_{\lambda,u}$, using Hölder's inequality, we have

$$\begin{aligned}\mathcal{I}_{\lambda,\mu}(w,z)&=\left(\frac{1}{2}-\frac{1}{2^*_s}\right)a(\|w\|^2+\|z\|^2)+\left(\frac{1}{4}-\frac{1}{2^*_s}\right)b(\|w\|^4+\|z\|^4)\\ &\quad-\left(\frac{1}{q}-\frac{1}{2^*_s}\right)\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx\right),\\ &\geq\left(\frac{1}{2}-\frac{1}{2^*_s}\right)a(\|w\|^2+\|z\|^2)+\left(\frac{1}{4}-\frac{1}{2^*_s}\right)b(\|w\|^4+\|z\|^4)\end{aligned}$$

$$-\left(\frac{1}{q}-\frac{1}{2_s^*}\right)(\kappa_s S(s,N))^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}}+(\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}(\|w\|^2+\|z\|^2)^{\frac{q}{2}}.$$

Since $2_s^*\geq 4$, the functional $\mathcal{I}_{\lambda,\mu}$ is coercive and hence bounded below in $\mathcal{N}_{\lambda,\mu}$. $\square$

Let

$$\Gamma_2:=(S(s,N))^{\frac{q}{2}}\left(\frac{S(s,\alpha,\beta)^{\frac{2_s^*}{2}}(2-q)}{2(2_s^*-2)}\right)^{\frac{2-q}{2_s^*-2}}\left(\frac{\kappa_s a(2_s^*-2)}{2_s^*-q}\right)^{\frac{2_s^*-q}{2_s^*-2}}. \tag{3.9}$$

Now we discuss the behavior of fibering maps with respect to the sign changing weights in the following two cases.

Case 1: $\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx>0$, for a fixed $(w,z)\in\mathcal{H}(\mathcal{C})$.

First we define $\psi_{(w,z)}:\mathbb{R}^+\to\mathbb{R}$ as

$$\begin{aligned}\psi_{(w,z)}(t)&=at^{2-2_s^*}(\|w\|^2+\|z\|^2)+bt^{4-2_s^*}(\|w\|^4+\|z\|^4)\\&\quad-t^{q-2_s^*}\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx\right).\end{aligned}$$

Observe that $(tw,tz)\in\mathcal{N}_{\lambda,\mu}$ if and only if

$$\psi_{(w,z)}(t)=2\int_\Omega |w(x,0)|^\alpha|z(x,0)|^\beta dx.$$

It is clear that $\lim_{t\to 0^+}\psi_{(w,z)}(t)=-\infty$ and $\lim_{t\to+\infty}\psi_{(w,z)}(t)=0^+$. Moreover,

$$\begin{aligned}\psi'_{(w,z)}(t)&=a(2-2_s^*)t^{1-2_s^*}(\|w\|^2+\|z\|^2)+b(4-2_s^*)t^{3-2_s^*}(\|w\|^4+\|z\|^4)\\&-(q-2_s^*)t^{q-2_s^*-1}\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx\right).\end{aligned}\tag{3.10}$$

Rewrite $\psi'_{(w,z)}(t)=t^{3-2_s^*}\mathcal{A}_{(w,z)}(t)$, where

$$\begin{aligned}\mathcal{A}_{(w,z)}(t)&=a(2-2_s^*)t^{-2}(\|w\|^2+\|z\|^2)+b(4-2_s^*)(\|w\|^4+\|z\|^4)\\&\quad-(q-2_s^*)t^{q-4}\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx\right).\end{aligned}$$

Then,

$$\begin{aligned}\mathcal{A}'_{(w,z)}(t)&=-2a(2-2_s^*)t^{-3}(\|w\|^2+\|z\|^2)\\&-(q-4)(q-2_s^*)t^{q-5}\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx\right).\end{aligned}$$

Therefore, it can be shown that there exists unique $t^*>0$ such that $\mathcal{A}'_{(w,z)}(t^*)=0$. Indeed,

$$t^*=\left(\frac{(2_s^*-q)(4-q)\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx\right)}{2a(2_s^*-2)(\|w\|^2+\|z\|^2)}\right)^{\frac{1}{2-q}}.$$

Furthermore, since $q<2$, $\lim_{t\to 0^+}\mathcal{A}'_{(w,z)}(t)=-\infty$ and $A'_{(w,z)}(t)>0$ for $t>>1$. Then, $\mathcal{A}'_{(w,z)}(t)<0$ for $t\in(0,t^*)$ and $\mathcal{A}'_{(w,z)}(t)>0$ for $t\in(t^*,+\infty)$. In addition,

$$\lim_{t\to 0^+}\mathcal{A}_{(w,z)}(t)=+\infty \ \text{ and } \lim_{t\to+\infty}\mathcal{A}_{(w,z)}(t)=b(4-2_s^*)(\|w\|^4+\|z\|^4)<0.$$

This implies that there exists a unique $0<t_*<t^*$ such that $\mathcal{A}_{(w,z)}(t_*)=0$. Therefore, from $\psi'_{(w,z)}(t)=t^{3-2_s^*}\mathcal{A}_{(w,z)}(t)$, we get t_* as a unique critical point of $\psi_{(w,z)}(t)$, which is the global maximum point.

Now we can estimate $\psi_{(w,z)}(t_*)$ from below as follows:

$$\psi_{(w,z)}(t)=\overline{\psi}_{(w,z)}(t)+bt^{4-2_s^*}(\|w\|^4+\|z\|^4)\geq\overline{\psi}_{(w,z)}(t),$$

where

$$\begin{aligned}\overline{\psi}_{(w,z)}(t)&=at^{2-2_s^*}(\|w\|^2+\|z\|^2)\\&\quad-t^{q-2_s^*}\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx\right).\end{aligned}$$

Using elementary calculus, we obtain

$$\max_{t>0}\overline{\psi}_{(w,z)}(t)=\overline{\psi}_{(w,z)}(t_c)$$

with

$$t_c=\left(\frac{(2_s^*-q)\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx+\mu\int_\Omega g(x)|z(x,0)|^q dx\right)}{(2_s^*-2)a(\|w\|^2+\|z\|^2)}\right)^{\frac{1}{2-q}}.$$

Therefore,

$$\begin{aligned}\psi_{(w,z)}(t_*)&\geq\overline{\psi}_{(w,z)}(t_c)\\&\geq\frac{\left(\frac{2-q}{2_s^*-2}\right)\left(\frac{a(2_s^*-2)}{2_s^*-q}\right)^{\frac{2_s^*-q}{2-q}}(\|w\|^2+\|z\|^2)^{\frac{2_s^*}{2}}}{\left((\kappa_s S(s,N))^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}}+(\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}\right)^{\frac{2_s^*-2}{2-q}}}.\end{aligned}$$

Now if

$$\begin{aligned}&\frac{\left(\frac{2-q}{2_s^*-2}\right)\left(\frac{a(2_s^*-2)}{2_s^*-q}\right)^{\frac{2_s^*-q}{2-q}}(\|w\|^2+\|z\|^2)^{\frac{2_s^*}{2}}}{\left((\kappa_s S(s,N))^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}}+(\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}\right)^{\frac{2_s^*-2}{2-q}}}\\&\quad>2\int_\Omega|w(x,0)|^\alpha|z(x,0)|^\beta dx\end{aligned}$$

or, equivalently, if $(\lambda,\mu)\in\mathcal{M}_{\Gamma_2}$, where Γ_2 is given in (3.9), then the equation

$$\psi_{(w,z)}(t)=2\int_\Omega|w(x,0)|^\alpha|z(x,0)|^\beta dx.$$

has exactly two positive solutions in t, that is, if $(\lambda,\mu) \in \mathcal{M}_{\Gamma_2}$, there exist a unique $t^+ = t^+((w,z)) < t_*$ and a unique $t^- = t^-((w,z)) > t_*$, such that

$$\psi_{(w,z)}(t^+) = 2\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx = \psi_{(w,z)}(t^-)$$

which implies (t^+w, t^+z) and $(t^-w, t^-z) \in \mathcal{N}_{\lambda,\mu}$. Besides, $\psi'_{(w,z)}(t^+) > 0$ and $\psi'_{(w,z)}(t^-) < 0$. Indeed, $\psi'_{(w,z)} > 0$ on $(0,t_*)$ and $\psi'_{(w,z)} < 0$ on $(t_*,+\infty)$. Then $(t^+w,t^+z) \in \mathcal{N}^+_{\lambda,\mu}$ and $(t^-w,t^-z) \in \mathcal{N}^-_{\lambda,\mu}$. Since

$$\phi'_{(w,z)}(t) = t^{2^*_s}(\psi_{(w,z)}(t) - \int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx),$$

we also get $\phi'_{(w,z)}(t) < 0$ for all $t \in [0,t^+)$ and $\phi'_{(w,z)}(t) > 0$ for all $t \in (t^+,t^-)$. Thus $\mathcal{I}_{\lambda,\mu}(t^+w,t^+z) = \min_{0\le t\le t^-} \mathcal{I}_{\lambda,\mu}(tw,tz)$. Also $\phi'_{(w,z)}(t) > 0$ for all $t \in [t^+,t^-)$, $\phi'_{(w,z)}(t^-) = 0$, and $\phi'_{(w,z)}(t) < 0$ for all $t \in (t^-,\infty)$ implies that $\mathcal{I}_{\lambda,\mu}(t^-w,t^-z) = \max_{t\ge t_*} \mathcal{I}_{\lambda,\mu}(tw,tz)$.

Case 2: $\lambda \int_\Omega f(x)|w(x,0)|^q dx + \mu \int_\Omega g(x)|z(x,0)|^q dx \le 0$.

Note that $\lim_{t\to+\infty} \psi_{(w,z)}(t) = 0^+$ and $\lim_{t\to 0^+} \psi_{(w,z)}(t) = +\infty$. Additionally, $\psi_{(w,z)}(t) \ge 0$ for all $t \ge 0$ and is decreasing. Hence for all $\lambda,\mu > 0$ there exists $\hat{t} > 0$ such that $(\hat{t}w,\hat{t}z) \in \mathcal{N}_{\lambda,\mu}$. Moreover, as $\psi'_{(w,z)}(\hat{t}) < 0$, we have $(\hat{t}w,\hat{t}z) \in \mathcal{N}^-_{\lambda,\mu}$ and $\mathcal{I}_{\lambda,\mu}(\hat{t}w,\hat{t}z) = \max_{t\ge 0} \mathcal{I}_{\lambda,\mu}(tw,tz)$. Thus from the discussion above we have the following lemma.

Lemma 3.4. (i) *If* $\lambda \int_\Omega f(x)|w(x,0)|^q dx + \mu \int_\Omega g(x)|z(x,0)|^q dx > 0$ *and* $(\lambda,\mu) \in \mathcal{M}_{\Gamma_2}$ *(defined in* (1.1) *and in* (3.9)*), there exist unique* $t^+((w,z)) < t_* < t^-((w,z))$ *such that* $(t^\pm w, t^\pm z) \in \mathcal{N}^\pm_{\lambda,\mu}$ *and*

$$\mathcal{I}_{\lambda,\mu}(t^+w,t^+z) = \min_{0\le t\le t^-} \mathcal{I}_{\lambda,\mu}(tw,tz), \quad \mathcal{I}_{\lambda,\mu}(t^-w,t^-z) = \max_{t\ge t_*} \mathcal{I}_{\lambda,\mu}(tw,tz).$$

(ii) *If* $\lambda \int_\Omega f(x)|w(x,0)|^q dx + \mu \int_\Omega g(x)|z(x,0)|^q dx \le 0$*, there exists a unique* $\hat{t}((w,z)) > 0$ *such that* $(\hat{t}w,\hat{t}z) \in \mathcal{N}^-_{\lambda,\mu}$ *and* $\mathcal{I}_{\lambda,\mu}(\hat{t}w,\hat{t}z) = \max_{t\ge 0} \mathcal{I}_{\lambda,\mu}(tw,tz)$.

The behavior of $\psi_{(w,z)}(t)$ and $\phi_{(w,z)}(t)$ in different cases is shown in Figure 1. We then deduce

Lemma 3.5. *Let* $(\lambda,\mu) \in \mathcal{M}_{\Gamma_2}$*, where* Γ_2 *is defined in* (3.9)*. Then* $\Theta_{\lambda,\mu} \le \Theta^+_{\lambda,\mu} < 0$.

Proof. Let $(w,z) \in \mathcal{H}(\mathcal{C})$ be such that

$$\lambda \int_\Omega f(x)|w(x,0)|^q dx + \mu \int_\Omega g(x)|z(x,0)|^q dx > 0.$$

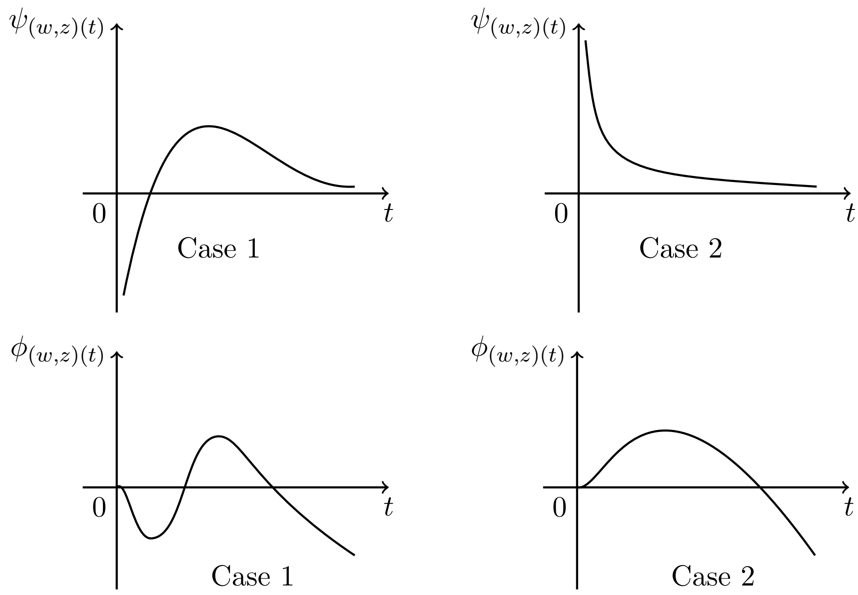

FIGURE 1.

Then by Lemma 3.4, for $(\lambda, \mu) \in \mathcal{M}_{\Gamma_2}$, there exists $t^+((w,z)) > 0$ such that $(t^+w, t^+z) \in \mathcal{N}^+_{\lambda,\mu}$. Denote $t^+w = w_1$ and $t^+z = z_1$. Therefore,

$$\mathcal{I}_{\lambda,\mu}(w_1, z_1) = \left(\frac{1}{2} - \frac{1}{q}\right) a(\|w_1\|^2 + \|z_1\|^2) + \left(\frac{1}{4} - \frac{1}{q}\right) b(\|w_1\|^4 + \|z_1\|^4)$$
$$+ 2\left(\frac{1}{q} - \frac{1}{2_s^*}\right) \int_\Omega |w_1(x,0)|^\alpha |z_1(x,0)|^\beta dx.$$

Since $(w_1, z_1) \in \mathcal{N}^+_{\lambda,\mu}$ and $2_s^* \geq 4$, we get

$$\mathcal{I}_{\lambda,\mu}(w_1, z_1) \leq -\frac{(2-q)(2_s^*-2)}{2(2_s^*)q} a(\|w_1\|^2 + \|z_1\|^2)$$
$$- \frac{(4-q)(2_s^*-4)}{4(2_s^*)q} b(\|w_1\|^4 + \|z_1\|^4) < 0. \qquad \square$$

Regarding the decomposition, $\mathcal{N}^-_{\lambda,\mu}$ of a Nehari manifold, we have the following lemma which helps us to show that the set $\mathcal{N}^-_{\lambda,\mu}$ is closed in the $\mathcal{H}(\mathcal{C})$-topology.

Lemma 3.6. *There exists $\delta > 0$ such that $\|(w,z)\| \geq \delta$ for all $(w,z) \in \mathcal{N}^-_{\lambda,\mu}$.*

Proof. Let $(w,z) \in \mathcal{N}^-_{\lambda,\mu}$ then from (3.3), we get

$$a(\|w\|^2 + \|z\|^2) + 3b(\|w\|^4 + \|z\|^4) - \lambda(q-1)\int_\Omega f(x)|w(x,0)|^q dx$$
$$- \mu(q-1)\int_\Omega g(x)|z(x,0)|^q dx < 2(2_s^*-1)\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx.$$

Now using (3.1) together with (2.5), we get

$$
\begin{aligned}
a(2-q)(\|w\|^2+\|z\|^2) &< 2(2_s^*-q)\int_\Omega |w(x,0)|^\alpha |z(x,0)|^\beta dx,\\
&< 2(2_s^*-q)(\kappa_s S(s,\alpha,\beta))^{\frac{-2_s^*}{2}}(\|w\|^2+\|z\|^2)^{\frac{2_s^*}{2}}
\end{aligned}
$$

which implies that $\|(w,z)\|^{2_s^*-2} > C$. Hence $\|(w,z)\| \geq \delta$ for some $\delta > 0$. □

Corollary 3.7. *Assume* $(\lambda,\mu) \in \mathcal{M}_{\Gamma_1}$, *where* Γ_1 *is defined in* (3.4)*. Then* $\mathcal{N}_{\lambda,\mu}^-$ *is closed set in the* $\mathcal{H}(\mathcal{C})$*-topology.*

Proof. Let $\{(w_k,z_k)\}$ be a sequence in $\mathcal{N}_{\lambda,\mu}^-$ such that $(w_k,z_k) \to (w,z)$ in $\mathcal{H}(\mathcal{C})$. Since $\mathcal{N}_{\lambda,\mu}^0 = \emptyset$ for $(\lambda,\mu) \in \mathcal{M}_{\Gamma_1}$, $(w,z) \in \overline{\mathcal{N}_{\lambda,\mu}^-} = \mathcal{N}_{\lambda,\mu}^- \cup \{0\}$. Now using Lemma 3.6, we get $\|(w,z)\| = \lim\limits_{k\to\infty} \|(w_k,z_k)\| \geq \delta > 0$. Hence $(w,z) \neq (0,0)$. Therefore $(w,z) \in \mathcal{N}_{\lambda,\mu}^-$. □

Now we note that the Nehari manifold lacks the basic vector space structure to talk about the derivative of the functional $\mathcal{I}_{\lambda,\mu}$ restricted to $\mathcal{N}_{\lambda,\mu}$. The following lemma is very crucial in this context, which helps us to construct a small *variation* of any point in $\mathcal{N}_{\lambda,\mu}$ which lies in the Nehari manifold. The proof follows from a direct application of the implicit function theorem.

Lemma 3.8. *For given* $(w,z) \in \mathcal{N}_{\lambda,\mu}$ *and* $(\lambda,\mu) \in \mathcal{M}_{\Gamma_1}$, *there exists* $\delta > 0$ *and a differentiable function* $\xi : \mathcal{B}(0,\delta) \subseteq \mathcal{H}(\mathcal{C}) \to \mathbb{R}$ *such that* $\xi(0) = 1$, *the function* $\xi(v_1,v_2)(w-v_1,z-v_2) \in \mathcal{N}_{\lambda,\mu}$ *and* $\langle \xi'(0),(v_1,v_2)\rangle_* = \frac{N}{D}$ *for all* $(v_1,v_2) \in \mathcal{H}(\mathcal{C})$, *where*

$$
\begin{aligned}
N = {} & 2a(\langle w,v_1\rangle + \langle z,v_2\rangle) + 4b(\|w\|^2\langle w,v_1\rangle + \|z\|^2\langle z,v_2\rangle)\\
& - q\lambda \int_\Omega f(x)|w(x,0)|^{q-2}w(x,0)v_1(x,0)dx\\
& - q\mu \int_\Omega g(x)|z(x,0)|^{q-2}z(x,0)v_2(x,0)dx)\\
& - 2\alpha \int_\Omega |w(x,0)|^{\alpha-2}w(x,0)|z(x,0)|^\beta v_1(x,0)dx\\
& - 2\beta \int_\Omega |w(x,0)|^\alpha |z(x,0)|^{\beta-2}z(x,0)v_2(x,0)dx,\\
D = {} & (2_s^*-2)a(\|w\|^2+\|z\|^2) + (2_s^*-4)b(\|w\|^4+\|z\|^4)\\
& - (2_s^*-q)\left(\lambda\int_\Omega f(x)|w(x,0)|^q dx + \mu\int_\Omega g(x)|z(x,0)|^q dx\right).
\end{aligned}
$$

Proof. The proof is quite standard. We argue as in Lemma 3.7 in [12] or Lemma 2.6 in [15]. Let

$$
\mathcal{F}_{(w,z)} : \mathbb{R}^+ \times \mathcal{H}(\mathcal{C}) \to \mathbb{R}
$$

defined by

$$
\begin{aligned}
\mathcal{F}_{(w,z)}(t,(v_1,v_2)) &= ta\|(v_1,v_2)-(w,z)\|^2 + t^3 b\|(v_1,v_2)-(w,z)\|^4 \\
&- t^{q-1}\lambda \int_\Omega f(x)|v_1(x,0)-w(x,0)|^q dx - t^{q-1}\mu \int_\Omega g(x)|v_2(x,0)-z(x,0)|^q dx \\
&- 2t^{2_s^*-1} \int_\Omega |v_1(x,0)-w(x,0)|^\alpha |v_2(x,0)-z(x,0)|^\beta dx,
\end{aligned}
$$

for any $(t,(v_1,v_2)) \in \mathbb{R}^+ \times \mathcal{H}(\mathcal{C})$. Observing that

$$
\mathcal{F}_{(w,z)}(1,(0,0)) = 0 \text{ and since } \mathcal{N}^0_{\lambda,\mu} = \emptyset,\ \frac{\partial F}{\partial t}(1,(0,0)) \neq 0,
$$

one can apply the implicit function theorem to get the result. □

Now using Lemma 3.8, we prove the following proposition which can help us to extract a Palais–Smale sequence out of Nehari decompositions using the Ekeland variational principle.

Proposition 3.9. (i) *Let $(\lambda,\mu) \in \mathcal{M}_{\Gamma_1} \cap \mathcal{M}_{\Gamma_2}$. Then there exists a minimizing sequence $\{(w_k,z_k)\} \subset \mathcal{N}_{\lambda,\mu}$ such that*

$$
\mathcal{I}_{\lambda,\mu}(w_k,z_k) = \Theta_{\lambda,\mu} + o_k(1) \ \textit{and}\ \|\mathcal{I}^{'}_{\lambda,\mu}(w_k,z_k)\|_* = o_k(1),
$$

where $\|\ \|_*$ is a norm of the dual space of $\mathcal{H}(\mathcal{C})$.

(ii) *Let $(\lambda,\mu) \in \mathcal{M}_{\Gamma_3}$ for some $\Gamma_3 > 0$. Then there exists a minimizing sequence $\{(w_k,z_k)\} \subset \mathcal{N}^-_{\lambda,\mu}$ such that*

$$
\mathcal{I}_{\lambda,\mu}(w_k,z_k) = \Theta^-_{\lambda,\mu} + o_k(1) \ \textit{and}\ \|\mathcal{I}^{'}_{\lambda,\mu}(w_k,z_k)\|_* = o_k(1).
$$

Proof. To avoid any repetition, we only prove the case (i) of the above Proposition. The proof for the case (ii) is similar. From Lemma 3.3, $\mathcal{I}_{\lambda,\mu}$ is bounded below on $\mathcal{N}_{\lambda,\mu}$. So by the Ekeland variational principle, there exists a minimizing sequence $\{(w_k,z_k)\} \in \mathcal{N}_{\lambda,\mu}$ such that

$$
\begin{aligned}
&\mathcal{I}_{\lambda,\mu}(w_k,z_k) \leq \Theta_{\lambda,\mu} + \frac{1}{k}, \\
&\mathcal{I}_{\lambda,\mu}(\bar{w},\bar{z}) \geq \mathcal{I}_{\lambda,\mu}(w_k,z_k) - \frac{1}{k}\|(\bar{w}-w_k,\bar{z}-z_k)\| \ \text{ for all } (\bar{w},\bar{z}) \in \mathcal{N}_{\lambda,\mu}.
\end{aligned} \tag{3.11}
$$

Using equation (3.11), it is easy to show that $(w_k,z_k) \not\equiv 0$ for k large. Indeed using (3.11), the fact that $(w_k,z_k) \in \mathcal{N}_{\lambda,\mu}$ and Lemma 3.5, we get for k large

$$
\begin{aligned}
\frac{\Theta_{\lambda,\mu}}{2} \geq \mathcal{I}_{\lambda,\mu}(w_k,z_k) &\geq \left(\frac{1}{2}-\frac{1}{2_s^*}\right) a(\|w_k\|^2+\|z_k\|^2) \\
&- \left(\frac{1}{q}-\frac{1}{2_s^*}\right)\left(\lambda\int_\Omega f(x)|w_k(x,0)|^q dx + \mu\int_\Omega g(x)|z_k(x,0)|^q dx\right)
\end{aligned} \tag{3.12}
$$

which implies

$$\begin{aligned}
-\frac{2_s^*q}{2}\Theta_{\lambda,\mu} &\leq \lambda(2_s^*-q)\left(\int_\Omega f(x)|w_k(x,0)|^q dx + \mu\int_\Omega g(x)|z_k(x,0)|^q dx\right)\\
&\leq (2_s^*-q)(\kappa_s S(s,N))^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}} + (\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}\|(w_k,z_k)\|^q.
\end{aligned} \tag{3.13}$$

From (3.13), we get immediately

$$\|(w_k,z_k)\| \geq \left(-\frac{2_s^*q}{2(2_s^*-q)}\Theta_{\lambda,\mu}(\kappa_s S(s,N))^{\frac{q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}} + (\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{q-2}{2}}\right)^{\frac{1}{q}}.$$

Besides this, combining (3.12) and (3.13), we obtain

$$\|(w_k,z_k)\| \leq \left(\frac{2(2_s^*-q)}{q(2_s^*-2)}(\kappa_s S(s,N))^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}} + (\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}\right)^{\frac{1}{2-q}}.$$

Next we claim that $\|\mathcal{I}'_{\lambda,\mu}(w_k,z_k)\|_* \to 0$ as $k\to\infty$. Now, using Lemma 3.8, we get the differentiable functions $\xi_k : \mathcal{B}(0,\delta_k) \to \mathbb{R}$ for some $\delta_k > 0$ such that $\xi_k(\bar{w},\bar{z})(w_k - \bar{w}, z_k - \bar{z}) \in \mathcal{N}_{\lambda,\mu}$, for all $(\bar{w},\bar{z}) \in \mathcal{B}(0,\delta_k)$. For fixed k, choose $0<\rho<\delta_k$. Let $(w,z) \in \mathcal{H}(\mathcal{C})$ with $(w,z) \not\equiv 0$ and let $(\bar{w},\bar{z})_\rho = \frac{\rho(w,z)}{\|(w,z)\|}$. We set $\eta_\rho = \xi_k((\bar{w},\bar{z})_\rho)((w_k,z_k) - (\bar{w},\bar{z})_\rho)$. Since $\eta_\rho \in \mathcal{N}_{\lambda,\mu}$, we get from (3.1)

$$\mathcal{I}_{\lambda,\mu}(\eta_\rho) - \mathcal{I}_{\lambda,\mu}(w_k,z_k) \geq -\frac{1}{k}\|\eta_\rho - (w_k,z_k)\|.$$

Now by the mean value theorem, we get

$$\left\langle \mathcal{I}'_{\lambda,\mu}(w_k,z_k), (\eta_\rho - (w_k,z_k))\right\rangle_* + o_k(\|(\eta_\rho - (w_k,z_k))\|) \geq -\frac{1}{k}\|(\eta_\rho - (w_k,z_k))\|.$$

Hence

$$\begin{aligned}
&\left\langle \mathcal{I}'_{\lambda,\mu}(w_k,z_k), -(\bar{w},\bar{z})_\rho\right\rangle_* + (\xi_k(\bar{w},\bar{z})_\rho - 1)\left\langle \mathcal{I}'_{\lambda,\mu}(w_k,z_k), (w_k,z_k) - (\bar{w},\bar{z})_\rho\right\rangle_*\\
&\geq -\frac{1}{k}\|(\eta_\rho - (w_k,z_k))\| + o(\|(\eta_\rho - (w_k,z_k))\|)
\end{aligned}$$

and since $\left\langle \mathcal{I}'_{\lambda,\mu}(\eta_\rho), ((w_k,z_k) - (\bar{w},\bar{z})_\rho)\right\rangle_* = 0$ by the fact that $\eta_\rho \in \mathcal{N}_{\lambda,\mu}$, we have

$$\begin{aligned}
-\rho\left\langle \mathcal{I}'_{\lambda,\mu}(w_k,z_k), \frac{(w,z)}{\|(w,z)\|}\right\rangle_* &+ (\xi_k((\bar{w},\bar{z})_\rho) - 1)\left\langle \mathcal{I}'_{\lambda,\mu}(w_k,z_k)\right.\\
&\left. - \mathcal{I}'_{\lambda,\mu}(\eta_\rho), ((w_k,z_k) - (\bar{w},\bar{z})_\rho)\right\rangle_*\\
&\geq -\frac{1}{k}\|(\eta_\rho - (w_k,z_k))\| + o(\|(\eta_\rho - (w_k,z_k))\|).
\end{aligned}$$

Thus

$$\begin{aligned}
&\left\langle \mathcal{I}'_{\lambda,\mu}(w_k,z_k), \frac{(w,z)}{\|(w,z)\|}\right\rangle_* \leq \frac{1}{k\rho}\|(\eta_\rho - (w_k,z_k))\| + \frac{o(\|(\eta_\rho - (w_k,z_k))\|)}{\rho}\\
&+ \frac{(\xi_k((\bar{w},\bar{z})_\rho) - 1)}{\rho}\left\langle \mathcal{I}'_{\lambda,\mu}(w_k,z_k) - \mathcal{I}'_{\lambda,\mu}(\eta_\rho), ((w_k,z_k) - (\bar{w},\bar{z})_\rho)\right\rangle_*.
\end{aligned} \tag{3.14}$$

Since $\|(\eta_\rho - (w_k, z_k))\| \leq \rho|\xi_k((\bar{w}, \bar{z})_\rho)| + |\xi_k((\bar{w}, \bar{z})_\rho) - 1|\|(w_k, z_k)\|$ and $\lim_{\rho \to 0^+} \dfrac{|\xi_k((\bar{w}, \bar{z})_\rho) - 1|}{\rho} \leq \|\xi_k'(0)\|_*$, taking limit $\rho \to 0^+$ in (3.14) and using (3.12) and (3.13), we get

$$\left\langle \mathcal{I}'_{\lambda,\mu}(w_k, z_k), \frac{(w, z)}{\|(w, z)\|} \right\rangle_* \leq \frac{C}{k}(1 + \|\xi_k'(0)\|_*)$$

for some constant $C > 0$, independent of (w, z). So if we can show that $\|\xi_k'(0)\|_*$ is bounded, then we are done. Now from Lemma 3.8, using the boundedness of $\{(w_k, z_k)\}$ and Hölder's inequality, for some $K > 0$, we get

$$\left\langle \xi_k'(0), (\bar{w}, \bar{z}) \right\rangle_* = \frac{K\|(\bar{w}, \bar{z})\|}{D},$$

where D is defined in Lemma 3.8. Therefore, to prove the claim, we only need to prove that the denominator in the above expression is bounded away from zero. Suppose not. Then there exists a subsequence, still denoted by $\{(w_k, z_k)\}$, such that

$$\begin{aligned}(2_s^* - 2)a(\|w_k\|^2 + \|z_k\|^2) &+ (2_s^* - 4)b(\|w_k\|^4 + \|z_k\|^4) \\ - (2_s^* - q)\left(\lambda \int_\Omega f(x)|w_k(x, 0)|^q dx + \mu \int_\Omega g(x)|z_k(x, 0)|^q dx\right) &= o_k(1). \end{aligned} \tag{3.15}$$

From Equations (3.15) and (3.7), we get $\mathcal{T}_{\lambda,\mu}(w_k, z_k) = o_k(1)$. Now following the proof of Lemma 3.2 we get $\mathcal{T}_{\lambda,\mu}(w_k, z_k) \geq C_1 > 0$ for all k, for some $C_1 > 0$ and $(\lambda, \mu) \in \mathcal{M}_{\Gamma_1}$, which gives a contradiction. □

4. Compactness of Palais–Smale sequences

Now we prove the following proposition which shows the compactness of Palais–Smale sequences at sub-critical energy levels:

Proposition 4.1. *Suppose $\{(w_k, z_k)\}$ is a sequence in $\mathcal{H}(\mathcal{C})$ such that*

$$\mathcal{I}_{\lambda,\mu}(w_k, z_k) \to c \text{ and } \mathcal{I}'_{\lambda,\mu}(w_k, z_k) \to 0,$$

where

$$\begin{aligned} c < c_{\lambda,\mu} &= \frac{2s}{N}\left(\frac{1}{2}a\kappa_s S(s, \alpha, \beta)\right)^{\frac{N}{2s}} \\ &- \left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}} + (\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2(2-q)}{4-q}} \frac{(4-q)(2_s^* - q)^{\frac{4}{4-q}}}{2(2_s^* q)} \left(\frac{(\kappa_s S(s, N))^{-2}}{b(2_s^* - 4)}\right)^{\frac{q}{4-q}} \end{aligned} \tag{4.1}$$

is a positive constant, then there exists a strongly convergent subsequence.

Proof. Let $\{(w_k, z_k)\}$ be a non-negative (one can assume without loss of any generality, since $\mathcal{I}_{\lambda,\mu}(|w|, |z|) = \mathcal{I}_{\lambda,\mu}(w, z)$) $(PS)_c$ sequence for $\mathcal{I}_{\lambda,\mu}$ in $\mathcal{H}(\mathcal{C})$, then it is easy to see that $\{(w_k, z_k)\}$ is bounded in $\mathcal{H}(\mathcal{C})$. Therefore there exists non-negative $(w_0, z_0) \in \mathcal{H}(\mathcal{C})$ such that, up to a subsequence,

$$\begin{gathered} w_k \rightharpoonup w_0 \text{ and } z_k \rightharpoonup z_0 \text{ in } L^{2_s^*}(\Omega \times \{0\}), \\ \|w_k\| \to \sigma_1 \text{ and } \|z_k\| \to \sigma_2, \text{ with } \sigma^2 = \sigma_1^2 + \sigma_2^2, \\ w_k \to w_0 \text{ and } z_k \to z_0 \text{ in } L^p(\Omega \times \{0\}) \text{ for any } p \in [1, 2_s^*), \\ w_k(x,0) \to w_0(x,0) \text{ and } z_k(x,0) \to z_0(x,0) \text{ a.e. in } \Omega \times \{0\}, \\ w_k \le h_1 \text{ and } z_k \le h_2 \text{ a.e. in } \Omega \times \{0\}, \end{gathered} \tag{4.2}$$

as $k \to \infty$, with $h_1, h_2 \in L^p(\Omega \times \{0\})$ for a fixed $p \in [1, 2_s^*)$. If $\sigma = 0$, we immediately see that $(w_k, z_k) \to (0,0)$ in $\mathcal{H}(\mathcal{C})$ as $k \to \infty$. Hence, let us assume that $\sigma > 0$.

By (4.2) and the Brezis–Lieb lemma [6, Theorem 2] (see also [11, Lemma 8] and [16, Lemma 2.1]), we have

$$\|w_k\|^2 = \|w_k - w_0\|^2 + \|w_0\|^2 + o_k(1), \tag{4.3}$$

$$\|z_k\|^2 = \|z_k - z_0\|^2 + \|z_0\|^2 + o_k(1), \tag{4.4}$$

$$\begin{aligned} \int_\Omega |w_k(x,0)|^\alpha |z_k(x,0)|^\beta dx = & \int_\Omega |w_k(x,0) - w_0(x,0)|^\alpha |z_k(x,0) - z_0(x,0)|^\beta dx \\ & + \int_\Omega |w_0(x,0)|^\alpha |z_0(x,0)|^\beta dx + o_k(1) \end{aligned} \tag{4.5}$$

as $k \to \infty$. Consequently, by $\mathcal{I}'_{\lambda,\mu}(w_k, z_k)(w_k - w_0, z_k - z_0) = 0$ as $k \to \infty$ together with (4.2), (4.3), (4.5) and **(fg)**, we get

$$\begin{aligned} & (a + b\sigma_1^2)(\sigma_1^2 - \|w_0\|^2) + (a + b\sigma_2^2)(\sigma_2^2 - \|z_0\|^2) \\ & - \frac{2\alpha}{\alpha+\beta} \int_\Omega |w_k(x,0)|^{\alpha-2} w_k(x,0) |z_k(x,0)|^\beta (w_k(x,0) - w_0(x,0)) dx \\ & - \frac{2\beta}{\alpha+\beta} \int_\Omega |w_k(x,0)|^\alpha |z_k(x,0)|^{\beta-2} z_k(x,0)(z_k(x,0) - z_0(x,0)) dx = o_k(1) \end{aligned}$$

which leads us us to

$$\begin{aligned} & (a + b\sigma_1^2) \lim_{k\to\infty} \|w_k - w_0\|^2 + (a + b\sigma_2^2) \lim_{k\to\infty} \|z_k - z_0\|^2 \\ & = 2 \lim_{k\to\infty} \int_\Omega |w_k(x,0) - w_0(x,0)|^\alpha |z_k(x,0) - z_0(x,0)|^\beta dx. \end{aligned} \tag{4.6}$$

Now, let us denote

$$\lim_{k\to\infty} \int_\Omega |w_k(x,0) - w_0(x,0)|^\alpha |z_k(x,0) - z_0(x,0)|^\beta dx = \ell^{2_s^*}. \tag{4.7}$$

Thus, from (4.6) and (4.7), we get the crucial formula

$$(a + b\sigma_1^2) \lim_{k\to\infty} \|w_k - w_0\|^2 + (a + b\sigma_2^2) \lim_{k\to\infty} \|z_k - z_0\|^2 = 2\ell^{2_s^*}. \tag{4.8}$$

Then, from (4.8), it is clear that $\ell \ge 0$. If $\ell = 0$, since $\sigma > 0$, by (4.8), we have $(w_k, z_k) \to (w_0, z_0)$ in $\mathcal{H}(\mathcal{C})$ as $k \to \infty$, concluding the proof. Thus, let us assume by contradiction that $\ell > 0$. By (2.5) and (4.7),

$$\|w_k - w_0\|^2 + \|z_k - z_0\|^2 \ge \kappa_s S(s, \alpha, \beta) \ell^2. \tag{4.9}$$

Using the notation in (4.2) and (4.8), together with (4.9), we get

$$2\ell^{2_s^*-2} \geq \kappa_s S(s,\alpha,\beta)(a+b\min(\sigma_1,\sigma_2)^2). \tag{4.10}$$

Note that (4.8) implies, in particular, that

$$(a+b\sigma_1^2)\big(\sigma_1^2-\|w_0\|^2\big)+(a+b\sigma_2^2)\big(\sigma_2^2-\|z_0\|^2\big)=2\ell^{2_s^*}. \tag{4.11}$$

Using (4.10) in (4.11) together with (4.9) it follows that

$$\begin{aligned}
\big(2\ell^{2_s^*}\big)^{\frac{2_s^*-2}{2}} &= \big((a+b\sigma_1^2)(\sigma_1^2-\|w_0\|^2)+(a+b\sigma_2^2)(\sigma_2^2-\|z_0\|^2)\big)^{\frac{2_s^*-2}{2}}\\
&\geq (a+b\min(\sigma_1,\sigma_2)^2)^{\frac{2_s^*-2}{2}}\big(\lim_{k\to\infty}\big(\|w_k-w_0\|^2+\|z_k-z_0\|^2\big)\big)^{\frac{2_s^*-2}{2}}\\
&\geq (a+b\min(\sigma_1,\sigma_2)^2)^{\frac{2_s^*-2}{2}}(\kappa_s S(s,\alpha,\beta))^{\frac{2_s^*-2}{2}}\ell^{2_s^*-2}\\
&\geq \frac{1}{2}(a+b\min(\sigma_1,\sigma_2)^2)^{\frac{2_s^*}{2}}(\kappa_s S(s,\alpha,\beta))^{\frac{2_s^*}{2}}.
\end{aligned}$$

From this, we obtain

$$\begin{aligned}
(\sigma^2)^{\frac{2_s^*-2}{2}} &\geq \big((\sigma_1^2-\|w_0\|^2)+(\sigma_2^2-\|z_0\|^2)\big)^{\frac{2_s^*-2}{2}}\\
&\geq \frac{1}{2}(\kappa_s S(s,\alpha,\beta))^{\frac{2_s^*}{2}}(a+b\min(\sigma_1,\sigma_2)^2)
\end{aligned} \tag{4.12}$$

so that we have

$$\begin{aligned}
\sigma^2 &\geq (\kappa_s S(s,\alpha,\beta))^{\frac{N}{2s}}\left(\frac{1}{2}(a+b\min(\sigma_1,\sigma_2)^2)\right)^{\frac{2}{2_s^*-2}}\\
&\geq (\kappa_s S(s,\alpha,\beta))^{\frac{N}{2s}}\left(\frac{a}{2}\right)^{\frac{2}{2_s^*-2}}.
\end{aligned} \tag{4.13}$$

Now,

$$\begin{aligned}
c &= \mathcal{I}_{\lambda,\mu}(w_k,z_k)-\frac{1}{2_s^*}\langle \mathcal{I}'_{\lambda,\mu}(w_k,z_k),(w_k,z_k)\rangle+o_k(1)\\
&= \left(\frac{1}{2}-\frac{1}{2_s^*}\right)a(\|w_k\|^2+\|z_k\|^2)+\left(\frac{1}{4}-\frac{1}{2_s^*}\right)b(\|w_k\|^4+\|z_k\|^4)\\
&\quad-\left(\frac{1}{q}-\frac{1}{2_s^*}\right)\left(\lambda\int_\Omega f(x)|w_k(x,0)|^q dx+\mu\int_\Omega g(x)|z_k(x,0)|^q dx\right)+o_k(1)\\
&\geq \left(\frac{1}{2}-\frac{1}{2_s^*}\right)a(\|w_k\|^2+\|z_k\|^2)+\left(\frac{1}{4}-\frac{1}{2_s^*}\right)b(\|w_k\|^4+\|z_k\|^4)\\
&\quad-\left(\frac{1}{q}-\frac{1}{2_s^*}\right)(\kappa_s S(s,N))^{\frac{-q}{2}}\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}}+(\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}}\times\\
&\qquad(\|w_k\|^q+\|z_k\|^q)+o_k(1).
\end{aligned}$$

Define

$$F_b(t) = \left(\frac{1}{4} - \frac{1}{2_s^*}\right) bt^4 \\ - \left(\frac{1}{q} - \frac{1}{2_s^*}\right) (\kappa_s S(s,N))^{\frac{-q}{2}} \left((\lambda \|f\|_\gamma)^{\frac{2}{2-q}} + (\mu \|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2-q}{2}} t^q$$

and using the critical point arguments, we obtain

$$F_b(t) \geq -\left((\lambda \|f\|_\gamma)^{\frac{2}{2-q}} + (\mu \|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2(2-q)}{4-q}} \\ \times \frac{(4-q)(2_s^*-q)^{\frac{4}{4-q}}}{4(2_s^* q)} \left(\frac{(\kappa_s S(s,N))^{-2}}{b(2_s^*-4)}\right)^{\frac{q}{4-q}}.$$

Hence using this estimate, we get

$$c + o_k(1) \geq \left(\frac{1}{2} - \frac{1}{2_s^*}\right) a(\|w_k\|^2 + \|z_k\|^2) \\ - \left((\lambda \|f\|_\gamma)^{\frac{2}{2-q}} + (\mu \|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2(2-q)}{4-q}} \frac{(4-q)(2_s^*-q)^{\frac{4}{4-q}}}{2(2_s^* q)} \left(\frac{(\kappa_s S(s,N))^{-2}}{b(2_s^*-4)}\right)^{\frac{q}{4-q}}.$$

Now, as $k \to \infty$, by (4.13), we have

$$c \geq \frac{2s}{N} \left(\frac{1}{2} a\kappa_s S(s,\alpha,\beta)\right)^{\frac{N}{2s}} \\ - \left((\lambda \|f\|_\gamma)^{\frac{2}{2-q}} + (\mu \|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2(2-q)}{4-q}} \frac{(4-q)(2_s^*-q)^{\frac{4}{4-q}}}{2(2_s^* q)} \left(\frac{(\kappa_s S(s,N))^{-2}}{b(2_s^*-4)}\right)^{\frac{q}{4-q}}$$

which contradicts the assumption $c < c_{\lambda,\mu}$, considering (4.1). This concludes the proof. □

5. Existence of the first solution

In this section we prove Theorem 1.1 (i), by a minimization argument on $\mathcal{N}_{\lambda,\mu}$. Note that for λ, μ small enough $\Theta_{\lambda,\mu} < 0$ from Lemma 3.5.

Proof of Theorem 1.1(i). Let us fix $\Gamma_4 > 0$ such that $c_{\lambda,\mu} > 0$ (defined in (4.1)) for $(\lambda,\mu) \in \mathcal{M}_{\Gamma_4}$. Set $\Gamma_0 = \min\{\Gamma_1, \Gamma_2, \Gamma_4\}$ (see definitions in (3.4) and (3.9)). Now as the functional $\mathcal{I}_{\lambda,\mu}$ is bounded below in $\mathcal{N}_{\lambda,\mu}$, we minimize $\mathcal{I}_{\lambda,\mu}$ in $\mathcal{N}_{\lambda,\mu}$. Using Proposition 3.9 (i), we get a minimizing Palais–Smale sequence such that $\mathcal{I}_{\lambda,\mu}(w_k, z_k) \to \Theta_{\lambda,\mu}$. Then it is easy to show that $\{(w_k, z_k)\}$ is bounded and therefore there exists $(w_0, z_0) \in \mathcal{H}(\mathcal{C})$ such that $(w_k, z_k) \rightharpoonup (w_0, z_0)$ in $\mathcal{H}(\mathcal{C})$. Now for $(\lambda,\mu) \in \mathcal{M}_\Gamma$, where $\Gamma \in (0, \Gamma_0)$ and using Lemma 3.5 and Proposition 4.1 we get that $(w_k, z_k) \to (w_0, z_0)$ in $\mathcal{H}(\mathcal{C})$ which implies that the pair (w_0, z_0) is a minimizer of $\mathcal{I}_{\lambda,\mu}$ in $\mathcal{N}_{\lambda,\mu}$ with $\mathcal{I}_{\lambda,\mu}(w_0, z_0) < 0$. Now we discuss some properties of this minimizer (w_0, z_0).

(i) $(w_0, z_0) \in \mathcal{N}^+_{\lambda,\mu}$ for $\Gamma \in (0, \Gamma_0)$.

Proof. If not then $(w_0, z_0) \in \mathcal{N}^-_{\lambda,\mu}$. Note that using $(w_0, z_0) \in \mathcal{N}_{\lambda,\mu}$ and $\mathcal{I}_{\lambda,\mu}(w_0, z_0) < 0$ we get

$$\lambda \int_\Omega f(x)|w_0(x,0)|^q dx + \mu \int_\Omega g(x)|z_0(x,0)|^q dx > 0.$$

Therefore from Lemma 3.4, we get unique $t^-((w_0, z_0)) > t^+((w_0, z_0)) > 0$ such that $t^-(w_0, z_0) \in \mathcal{N}^-_{\lambda,\mu}$ and $t^+(w_0, z_0) \in \mathcal{N}^+_{\lambda,\mu}$ which implies $t^- = 1$ and $t^+ < 1$. Since $t \to \mathcal{I}_{\lambda,\mu}(t(w_0, z_0))$ is strictly increasing on (t^+, t^-),

$$\begin{aligned}\mathcal{I}_{\lambda,\mu}(t^+(w_0, z_0)) &= \min_{0\le t\le t^-} \mathcal{I}_{\lambda,\mu}(t(w_0, z_0)) < \mathcal{I}_{\lambda,\mu}(t^-(w_0, z_0))\\ &= \mathcal{I}_{\lambda,\mu}(w_0, z_0) = \Theta_{\lambda,\mu}\end{aligned}$$

which is a contradiction. Hence $(w_0, z_0) \in \mathcal{N}^+_{\lambda,\mu}$. As $\mathcal{I}_{\lambda,\mu}(w, z) = \mathcal{I}_{\lambda,\mu}(|w|, |z|)$, we can assume that $w_0 \ge 0, z_0 \ge 0$. □

(ii) (w_0, z_0) is not semi-trivial, that is, $w_0 \not\equiv 0$, $z_0 \not\equiv 0$.

Proof. Suppose by contradiction that $z_0 \equiv 0$. Then w_0 is a non-trivial non-negative solution of

$$\begin{cases} -\mathrm{div}(y^{1-2s}\nabla w) = 0, & \text{in } \mathcal{C},\\ w = 0 & \text{on } \partial_L,\\ M(\|w\|^2)\frac{\partial w}{\partial \nu} = \lambda f(x)|w|^{q-2}w & \text{on } \Omega \times \{0\}, \end{cases}$$

By the maximum principle [8], we get $w_0 > 0$ in $E^1_0(\mathcal{C})$ and

$$M(\|w_0\|^2)\|w_0\|^2 = \lambda \int_\Omega f(x)|w_0(x,0)|^q dx > 0. \tag{5.1}$$

Moreover, we can choose $\tilde{z}_0 \in E^1_0(\mathcal{C})\backslash\{0\}$ such that

$$\int_\Omega g(x)|\tilde{z}_0(x,0)|^q dx > 0.$$

Since $0 < q < 2$, from the mean value theorem, there exists $t_0 > 0$ such that $z_0^* = t_0\tilde{z}_0$ satisfies

$$M(\|z_0^*\|^2)\|z_0^*\|^2 = \mu \int_\Omega g(x)|z_0^*(x,0)|^q dx > 0. \tag{5.2}$$

Therefore,

$$0 < a(\|w_0\|^2 + \|z_0^*\|^2) \le \lambda \int_\Omega f(x)|w_0(x,0)|^q dx + \mu \int_\Omega g(x)|z_0^*(x,0)|^q dx. \tag{5.3}$$

Hence by Lemma 3.4 there exist unique $0 < t^+ < t_* < t^-$ such that $(t^+w_0, t^+z_0^*) \in \mathcal{N}^+_{\lambda,\mu}$ and

$$\mathcal{I}_{\lambda,\mu}(t^+w_0, t^+z_0^*) = \inf_{0\le t\le t^-} \mathcal{I}_{\lambda,\mu}(tw_0, tz_0^*).$$

From (3.10), one can show that the positive zero t_* of $\psi'_{w,z}(t) = 0$ satisfies

$$t_* \ge \left(\frac{(2_s^* - q)\left(\lambda \int_\Omega f(x)|w_0(x,0)|^q dx + \mu \int_\Omega g(x)|z_0^*(x,0)|^q dx\right)}{a(2_s^* - 2)(\|w_0\|^2 + \|z_0^*\|^2)}\right)^{\frac{1}{2-q}}.$$

Using (5.3), it is easy to see that $t_* > 1$. Using (5.1) and (5.2) we have

$$\mathcal{I}_{\lambda,\mu}(w_0, z_0^*) < \mathcal{I}_{\lambda,\mu}(w_0, 0).$$

Therefore,

$$\Theta^+_{\lambda,\mu} \leq \mathcal{I}_{\lambda,\mu}(t^+ w_0, t^+ z_0^*) \leq \mathcal{I}_{\lambda,\mu}(w_0, z_0^*) < \mathcal{I}_{\lambda,\mu}(w_0, 0) = \Theta^+_{\lambda,\mu}$$

which is a contradiction. Hence $w_0 \not\equiv 0$ and $z_0 \not\equiv 0$. Now using the fact that $M(t)$ is positive for $t > 0$ and strong maximum principle (see Lemma 2.6 [8]), we get $w_0 > 0, z_0 > 0$. □

(iii) (w_0, z_0) is indeed a local minimizer of $\mathcal{I}_{\lambda,\mu}$ in $\mathcal{H}(\mathcal{C})$.

Proof. Since $(w_0, z_0) \in \mathcal{N}^+_{\lambda,\mu}$, we have $t^+ = 1 < t_*$. Hence by continuity of $(w, z) \mapsto t_*$, for $\ell > 0$ small enough, there exists $\delta = \delta(\ell) > 0$ such that $1+\ell < t_*(w_0-w, z_0-z)$ for all $\|(w, z)\| < \delta$. Also, from Lemma 3.8, for $\delta > 0$ small enough, we obtain a C^1 map $t : \mathcal{B}(0, \delta) \to \mathbb{R}^+$ such that $t(w_0 - w, z_0 - z) \in \mathcal{N}_{\lambda,\mu}$, $t(0) = 1$. Therefore, for $\ell > 0$ and $\delta = \delta(\ell) > 0$ small enough, we have $t^+(w_0 - w, z_0 - z) = t(w, z) < 1+\ell < t_*(w_0 - w, z_0 - z)$ for all $\|(w, z)\| < \delta$. Since $t_*(w_0 - w, z_0 - z) > 1$, we obtain $\mathcal{I}_{\lambda,\mu}(w_0, z_0) \leq \mathcal{I}_{\lambda,\mu}(t^+(w_0 - w, z_0 - z)) \leq \mathcal{I}_{\lambda,\mu}(w_0 - w, z_0 - z)$ for all $\|(w, z)\| < \delta$. This shows that (w_0, z_0) is a local minimizer for $\mathcal{I}_{\lambda,\mu}$ in $\mathcal{H}(\mathcal{C})$. □

This concludes the proof of Theorem 1.1(i). □

6. Existence of the second solution in $\mathcal{N}^-_{\lambda,\mu}$

Now we show the existence of a second solution in $\mathcal{N}^-_{\lambda,\mu}$. Suppose

$$\mathcal{P} = \{x \in \Omega \mid f(x) > 0\} \cap \{x \in \Omega \mid g(x) > 0\}$$

has a positive measure. Consider the test functions as $\eta \in C^\infty_c(\mathcal{C}_\mathcal{P})$, where $\mathcal{C}_\mathcal{P} = \mathcal{P} \times [0, \infty)$ such that $0 \leq \eta(x, y) \leq 1$ in $\mathcal{C}_\mathcal{P}$ and

$$(\text{supp } f^+ \times \{y > 0\}) \cap (\text{supp } g^+ \times \{y > 0\}) \cap \{(x, y) \in \mathcal{C}_\mathcal{P} : \eta = 1\} \neq \emptyset.$$

Moreover, for $\rho > 0$ small, $\eta(x, y) = 1$ on $\mathcal{B}_\rho(0)^+$ and $\eta(x, y) = 0$ on $\mathcal{B}^c_{2\rho}(0)$. We take ρ small enough such that $\mathcal{B}_{2\rho}(0) \subset \mathcal{C}_\mathcal{P}$. Consider $(w_{\epsilon,\eta,\alpha} = \eta\sqrt{\alpha} w_\epsilon, w_{\epsilon,\eta,\beta} = \eta\sqrt{\beta} w_\epsilon) \in \mathcal{H}(\mathcal{C})$, where $w_\epsilon = E_s(u_\epsilon)$ and u_ϵ is defined in (2.4). Then we have the following lemma.

Lemma 6.1. *Let (w_0, z_0) be the local minimum for the functional $\mathcal{I}_{\lambda,\mu}$ in $\mathcal{H}(\mathcal{C})$ proved in the previous section and let $\frac{N-2s}{2} < p < N - 2s$. Then there exist $b_0 > 0$ small enough such that for any $0 < b < b_0$ and a suitable $\Gamma_* := \Gamma_*(b) > 0$, we have for $\epsilon = b^{1/p}$ and $(\lambda, \mu) \in \mathcal{M}_{\Gamma_*}$*

$$\mathcal{I}_\lambda(w_0 + r\, w_{\epsilon,\eta,\alpha}, z_0 + r\, w_{\epsilon,\eta,\beta}) < c_{\lambda,\mu}, \quad \forall r > 0.$$

Proof. Using the definition of $\mathcal{I}_{\lambda,\mu}$, we open the proof as follows.

$$\mathcal{I}_{\lambda,\mu}(w_0 + r\ w_{\epsilon,\eta,\alpha}, z_0 + r\ w_{\epsilon,\eta,\beta}) = \frac{a}{2}(\|w_0 + r\ w_{\epsilon,\eta,\alpha}\|^2 + \|z_0 + r\ w_{\epsilon,\eta,\beta}\|^2)$$
$$+ \frac{b}{4}(\|w_0 + r\ w_{\epsilon,\eta,\alpha}\|^4 + \|z_0 + r\ w_{\epsilon,\eta,\beta}\|^4) - \frac{\lambda}{q}\int_\Omega f(x)|w_0 + r\ w_{\epsilon,\eta,\alpha}|^q dx$$
$$- \frac{\mu}{q}\int_\Omega g(x)|z_0 + r\ w_{\epsilon,\eta,\beta}|^q dx - \frac{2}{2_s^*}\int_\Omega |w_0 + r\ w_{\epsilon,\eta,\alpha}|^\alpha |z_0 + r\ w_{\epsilon,\eta,\beta}|^\beta dx.$$

Using the fact that (w_0, z_0) is a positive solution of problem $(S_{\lambda,\mu})$, we get

$$\mathcal{I}_{\lambda,\mu}(w_0 + r\ w_{\epsilon,\eta,\alpha}, z_0 + r\ w_{\epsilon,\eta,\beta}) \leq \mathcal{I}_{\lambda,\mu}(w_0, z_0) + \frac{a}{2}r^2(\|w_{\epsilon,\eta,\alpha}\|^2 + \|w_{\epsilon,\eta,\beta}\|^2)$$
$$+ \frac{b}{4}r^4(\|w_{\epsilon,\eta,\alpha}\|^4 + \|w_{\epsilon,\eta,\beta}\|^4) + \frac{3}{2}br^2(\|w_0\|^2\|w_{\epsilon,\eta,\alpha}\|^2 + \|z_0\|^2\|w_{\epsilon,\eta,\beta}\|^2)$$
$$+ b\ r^3(\|w_0\|\|w_{\epsilon,\eta,\alpha}\|^3 + \|z_0\|\|w_{\epsilon,\eta,\beta}\|^3)$$
$$- \frac{\lambda}{q}\int_\Omega f(x)(|w_0 + r\ w_{\epsilon,\eta,\alpha}|^q - |w_0|^q - qr|w_0|^{q-1}w_{\epsilon,\eta,\alpha})(x,0)dx$$
$$- \frac{\mu}{q}\int_\Omega g(x)(|z_0 + r\ w_{\epsilon,\eta,\beta}|^q - |z_0|^q - qr|z_0|^{q-1}w_{\epsilon,\eta,\beta})(x,0)dx$$
$$- \frac{2}{2_s^*}\int_\Omega \Big(|w_0 + rw_{\epsilon,\eta,\alpha}|^\alpha |z_0 + rw_{\epsilon,\eta,\beta}|^\beta - |w_0|^\alpha |z_0|^\beta$$
$$-\alpha r|w_0|^{\alpha-1}|z_0|^\beta w_{\epsilon,\eta,\alpha} - \beta r|w_0|^\alpha |z_0|^{\beta-1}w_{\epsilon,\eta,\beta}\Big)(x,0)dx.$$

Note that supp $(w_{\epsilon,\eta,\alpha})$ = supp $(w_{\epsilon,\eta,\beta}) \subset \mathcal{C}_\mathcal{P}$ and $f(x), g(x) > 0$ for $(x,0) \in \mathcal{C}_\mathcal{P}$. Now based on

$$(a+b)^t \geq a^t + b^t + ta^{t-1}b + C_1 ab^{t-1},\ a, b > 0,\ t \geq 2, \tag{6.1}$$

we use the inequality

$$(m+n)^s(p+q)^t \geq m^s p^t + n^s q^t + sm^{s-1}np^t + tm^s p^{t-1}q + C_2 mn^{s-1}q^t + C_3 q^{t-1}n^s p,$$

for some non-negative constants C_1, C_2, C_3 and $p, q, m, n \geq 0,\ s, t \geq 2$ (see Appendix for the proof of (6.1)). Now using the above inequality together with Lemma 3.5 and Young's inequality we get the

$$\mathcal{I}_{\lambda,\mu}(w_0 + r\ w_{\epsilon,\eta,\alpha}, z_0 + r\ w_{\epsilon,\eta,\beta}) \leq \frac{a}{2}r^2(\|w_{\epsilon,\eta,\alpha}\|^2 + \|w_{\epsilon,\eta,\beta}\|^2)$$
$$+ \frac{7}{4}br^4(\|w_{\epsilon,\eta,\alpha}\|^4 + \|w_{\epsilon,\eta,\beta}\|^4) - \frac{2}{2_s^*}r^{2_s^*}\int_\Omega |w_{\epsilon,\eta,\alpha}(x,0)|^\alpha |w_{\epsilon,\eta,\beta}(x,0)|^\beta dx$$
$$- C_5 r^{2_s^*-1}\int_\Omega |w_\epsilon(x,0)|^{2_s^*-1}dx + 2C_4 b,$$

where $C_4 = (\|w_0\|^4 + \|z_0\|^4)$ and C_5 is obtained on combining C_2 and C_3. Now assume

$$\begin{aligned}G(t) &= \frac{a}{2}t^2(\|w_{\epsilon,\eta,\alpha}\|^2 + \|w_{\epsilon,\eta,\beta}\|^2) + \frac{7}{4}bt^4(\|w_{\epsilon,\eta,\alpha}\|^4 + \|w_{\epsilon,\eta,\beta}\|^4)\\ &\quad - \frac{2}{2_s^*}t^{2_s^*}\int_\Omega |w_{\epsilon,\eta,\alpha}(x,0)|^\alpha |w_{\epsilon,\eta,\beta}(x,0)|^\beta dx - C_5 t^{2_s^*-1}\int_\Omega |w_\epsilon(x,0)|^{2_s^*-1}dx.\end{aligned}$$

Since $\lim\limits_{t\to+\infty} G(t) = -\infty$ and $\lim\limits_{t\to 0^+} G(t) > 0$, there exists $t_\epsilon > 0$ such that

$$G(t_\epsilon) = \sup_{t\geq 0} g(t) \text{ and } \frac{d}{dt}G(t)\mid_{t=t_\epsilon} = 0. \tag{6.2}$$

From (6.2), we get

$$\begin{aligned}&at_\epsilon(\|w_{\epsilon,\eta,\alpha}\|^2 + \|w_{\epsilon,\eta,\beta}\|^2) + 7bt_\epsilon^3(\|w_{\epsilon,\eta,\alpha}\|^4 + \|w_{\epsilon,\eta,\beta}\|^4)\\ &= 2t_\epsilon^{2_s^*-1}\int_\Omega |w_{\epsilon,\eta,\alpha}(x,0)|^\alpha |w_{\epsilon,\eta,\beta}(x,0)|^\beta dx + C_6 t_\epsilon^{2_s^*-2}\int_\Omega |w_\epsilon(x,0)|^{2_s^*-1}dx.\end{aligned} \tag{6.3}$$

From (6.3), it is clear that t_ϵ is bounded from below. Moreover,

$$\begin{aligned}&\frac{a}{t_\epsilon^2}(\|w_{\epsilon,\eta,\alpha}\|^2 + \|w_{\epsilon,\eta,\beta}\|^2) + 7b(\|w_{\epsilon,\eta,\alpha}\|^4 + \|w_{\epsilon,\eta,\beta}\|^4)\\ &= 2t_\epsilon^{2_s^*-4}\int_\Omega |w_{\epsilon,\eta,\alpha}(x,0)|^\alpha |w_{\epsilon,\eta,\beta}(x,0)|^\beta dx + C_6 t_\epsilon^{2_s^*-5}\int_\Omega |w_\epsilon(x,0)|^{2_s^*-1}dx.\end{aligned} \tag{6.4}$$

Since $2_s^* \geq 4$, t_ϵ is bounded from above as well. If not, then the left-hand side of (6.4) is bounded for large values of t_ϵ, while the right-hand side goes to $+\infty$ as $t \to +\infty$, which is a contradiction. Hence from the above arguments, there exist positive constants t_1, t_2 such that $0 < t_1 \leq t_\epsilon \leq t_2 < \infty$.

Now from [3, 22] we get that the family $\{w_\epsilon\}$ and its trace on $\{y = 0\}$ satisfy

$$\begin{aligned}\|\eta w_\epsilon\|^2 &= \|w_\epsilon\|^2 + O(\epsilon^{N-2s}),\\ \int_\Omega |\eta w_\epsilon(x,0)|^{2_s^*}dx &= \int_{\mathbb{R}^N}\frac{1}{(1+|x|^2)^N}dx + O(\epsilon^N),\\ \frac{\|\eta w_\epsilon\|^2}{\|\eta w_\epsilon\|_{2_s^*}^2} &= \kappa_s S(s,N) + O(\epsilon^{N-2s}),\\ \int_\Omega |\eta w_\epsilon(x,0)|^{2_s^*-1} &\geq C_7\epsilon^{\frac{N-2s}{2}}dx.\end{aligned} \tag{6.5}$$

Using the estimates of (6.5), we estimate the second and fourth term in $G(t)$ as

$$\begin{aligned}&\frac{7}{4}bt^4(\|w_{\epsilon,\eta,\alpha}\|^4 + \|w_{\epsilon,\eta,\beta}\|^4) - C_5 t^{2_s^*-1}\int_\Omega |w_\epsilon(x,0)|^{2_s^*-1}dx\\ &\leq \frac{7}{4}bt_2^4(\|w_{\epsilon,\eta,\alpha}\|^4 + \|w_{\epsilon,\eta,\beta}\|^4) - C_5 t_1^{2_s^*-1}\int_\Omega |w_\epsilon(x,0)|^{2_s^*-1}dx\\ &\leq C_8 b(S(s,N)^{\frac{2N}{s}} + O(\epsilon^{N-2s})) - C_9\epsilon^{\frac{N-2s}{2}}.\end{aligned}$$

Now since $\epsilon = b^{1/p}$, with $N-2s > p > \frac{N-2s}{2}$, we get

$$\begin{aligned}&\frac{7}{4}bt^4(\|w_{\epsilon,\eta,\alpha}\|^4 + \|w_{\epsilon,\eta,\beta}\|^4) - C_5 t^{2_s^*-1}\int_\Omega |w_\epsilon(x,0)|^{2_s^*-1}dx\\ &\leq C_{10}\epsilon^p + C_{11}\epsilon^{N+p-2s} - C_9\epsilon^{\frac{N-2s}{2}}.\end{aligned}$$

Using these estimates we get

$$\begin{aligned}&\sup_{t\geq 0} G(t) = G(t_\epsilon) \leq \left(a(\alpha+\beta)t_\epsilon^2\|\eta w_\epsilon\|^2 - \frac{2}{2_s^*}t_\epsilon^{2_s^*}\int_\Omega \alpha^{\frac{\alpha}{2}}\beta^{\frac{\beta}{2}}|\eta w_\epsilon(x,0)|^{2_s^*}dx\right)\\ &\quad + C_{10}\epsilon^p + C_{11}\epsilon^{N+p-2s} - C_9\epsilon^{\frac{N-2s}{2}}\\ &\leq \left(\frac{1}{2}-\frac{1}{2_s^*}\right)\left(\frac{1}{2}\right)^{\frac{2}{2_s^*-2}}(a\kappa_s)^{\frac{2_s^*}{2_s^*-2}}\left[\left(\frac{\alpha}{\beta}\right)^{\frac{\beta}{\alpha+\beta}}+\left(\frac{\beta}{\alpha}\right)^{\frac{\alpha}{\alpha+\beta}}\right]^{\frac{2_s^*}{2_s^*-2}} S(s,N)^{\frac{2_s^*}{2_s^*-2}}\\ &\quad + O(\epsilon^{N-2s}) + C_{10}\epsilon^p + C_{11}\epsilon^{N+p-2s} - C_9\epsilon^{\frac{N-2s}{2}},\end{aligned}$$

where $C_9, C_{10}, C_{11} > 0$ are positive constants independent of ϵ, λ. Now using Lemma 2.5, we get that

$$\begin{aligned}&\mathcal{I}_{\lambda,\mu}(w_0 + r\ w_{\epsilon,\eta,\alpha}, z_0 + r\ w_{\epsilon,\eta,\beta}) \leq \frac{2s}{N}\left(\frac{1}{2}(a\kappa_s S(s,\alpha,\beta))\right)^{\frac{N}{2s}}\\ &\quad + C_{12}\epsilon^{N-2s} + C_{10}\epsilon^p + C_{11}\epsilon^{N+p-2s} - C_9\epsilon^{\frac{N-2s}{2}} + 2\epsilon^p R^4\\ &\leq \frac{2s}{N}\left(\frac{1}{2}(a\kappa_s S(s,\alpha,\beta))\right)^{\frac{N}{2s}} + C_{12}\epsilon^{N-2s} + C_{13}\epsilon^p + C_{11}\epsilon^{N+p-2s} - C_9\epsilon^{\frac{N-2s}{2}}.\end{aligned}$$

Observe that there exists $\epsilon_0 > 0$ sufficiently small such that

$$C_{12}\epsilon^{N-2s} + C_{13}\epsilon^p + C_{11}\epsilon^{N+p-2s} - C_9\epsilon^{\frac{N-2s}{2}} \leq C_{14}\epsilon^p - C_9\epsilon^{\frac{N-2s}{2}} < 0$$

for $\epsilon \in (0,\epsilon_0)$. Hence for $\epsilon \in (0,\epsilon_0)$ (and consequently $b \in (0,b_0)$ with $b_0 := \epsilon_0^p$), there exist $\lambda, \mu > 0$ satisfying the inequality

$$\begin{aligned}&-\left((\lambda\|f\|_\gamma)^{\frac{2}{2-q}} + (\mu\|g\|_\gamma)^{\frac{2}{2-q}}\right)^{\frac{2(2-q)}{4-q}}\frac{(4-q)(2_s^*-q)^{\frac{4}{4-q}}}{2(2_s^*q)}\left(\frac{(\kappa_s S(s,N))^{-2}}{(2_s^*-4)}\right)^{\frac{q}{4-q}}\\ &\geq b^{\frac{q}{4-q}}(C_{14}\epsilon^p - C_9\epsilon^{\frac{N-2s}{2}}).\end{aligned}$$

Equivalently, for $0 < b < b_0$, if we choose $(\lambda,\mu) \in \mathcal{M}_{\Gamma_*}$, where

$$\Gamma_* = \Gamma_*(b) = \left(\frac{\kappa_s^2 S(s,N)^2(2_s^*-4)}{2_s^*-q}\right)^{\frac{q}{4}}\left(\frac{2(2_s^*q)b^{\frac{q}{4}}(C_9 b^{\frac{N-2s}{2p}} - C_{14}b)}{4-q}\right)^{\frac{4-q}{4}},$$

then $\mathcal{I}_{\lambda,\mu}(w_0 + r\ w_{\epsilon,\eta,\alpha}, z_0 + r\ w_{\epsilon,\eta,\beta}) < c_{\lambda,\mu}$ for all $r > 0$. This proves the lemma. $\square$

Now consider

$$W_1 = \left\{(w,z) \in \mathcal{H}(\mathcal{C}) \setminus \{0\} \Big| \frac{1}{\|(w,z)\|} t^- \left(\frac{(w,z)}{\|(w,z)\|} \right) > 1 \right\} \cup \{0\},$$
$$W_2 = \left\{(w,z) \in \mathcal{H}(\mathcal{C}) \setminus \{0\} \Big| \frac{1}{\|(w,z)\|} t^- \left(\frac{(w,z)}{\|(w,z)\|} \right) < 1 \right\}.$$

Then $\mathcal{N}^-_{\lambda,\mu}$ disconnects $\mathcal{H}(\mathcal{C})$ in two connected components W_1 and W_2 and $\mathcal{H}(\mathcal{C}) \setminus \mathcal{N}^-_{\lambda,\mu} = W_1 \cup W_2$. For each $(w,z) \in \mathcal{N}^+_{\lambda,\mu}$, we have $1 < t_{\max}((w,z)) < t^-((w,z))$. Since $t^-((w,z)) = \frac{1}{\|(w,z)\|} t^- \left(\left(\frac{(w,z)}{\|(w,z)\|} \right) \right)$, then $\mathcal{N}^+_{\lambda,\mu} \subset W_1$. In particular, $(w_0, z_0) \in W_1$. Now we claim that there exists $l_0 > 0$ such that $(w_0 + l_0 w_{\epsilon,\eta}, z_0 + l_0 w_{\epsilon,\eta}) \in W_2$ with $\epsilon = b^{1/p}$ and $b < b_0$ given in Lemma 6.1.

First, we find a constant $c > 0$ such that

$$0 < t^- \left(\left(\frac{(w_0 + l\, w_{\epsilon,\eta}, z_0 + l\, w_{\epsilon,\eta})}{\|(w_0 + l\, w_{\epsilon,\eta}, z_0 + l\, w_{\epsilon,\eta})\|} \right) \right) < c, \quad \forall\, l > 0.$$

Otherwise, there exists a sequence $\{l_k\}$ such that, as $k \to \infty$, $l_k \to \infty$ and

$$t^- \left(\left(\frac{(w_0 + l_k\, w_{\epsilon,\eta}, z_0 + l_k\, w_{\epsilon,\eta})}{\|(w_0 + l_k\, w_{\epsilon,\eta}, z_0 + l_k\, w_{\epsilon,\eta})\|} \right) \right) \to \infty.$$

Let

$$(\bar{w}_k, \bar{z}_k) = \frac{(w_0 + l_k\, w_{\epsilon,\eta}, z_0 + l_k\, w_{\epsilon,\eta})}{\|(w_0 + l_k\, w_{\epsilon,\eta}, z_0 + l_k\, w_{\epsilon,\eta})\|}.$$

Since $t^-((\bar{w}_k, \bar{z}_k))(\bar{w}_k, \bar{z}_k) \in \mathcal{N}^-_{\lambda,\mu} \subset \mathcal{N}_{\lambda,\mu}$ and by the Lebesgue dominated convergence theorem,

$$\begin{aligned}
&\lim_{k\to\infty} \int_\Omega |\bar{w}_k(x,0)|^\alpha |\bar{z}_k(x,0)|^\beta dx \\
&\quad = \lim_{k\to\infty} \frac{\int_\Omega |(w_0 + l_k\, w_{\epsilon,\eta})(x,0)|^\alpha (z_0 + l_k\, w_{\epsilon,\eta})(x,0)|^\beta dx}{\|(w_0 + l_k\, w_{\epsilon,\eta}, z_0 + l_k\, w_{\epsilon,\eta})\|^{2^*_s}} \\
&\quad = \lim_{k\to\infty} \frac{\int_\Omega |(w_0/l_k + w_{\epsilon,\eta})(x,0)|^\alpha |(z_0/l_k + w_{\epsilon,\eta})(x,0)|^\beta dx}{\|(w_0/l_k + \ w_{\epsilon,\eta}, z_0/l_k + \ w_{\epsilon,\eta})\|^{2^*_s}} \\
&\quad = \frac{\int_\Omega (w_{\epsilon,\eta}(x,0))^{2^*_s} dx}{\|(w_{\epsilon,\eta}, w_{\epsilon,\eta})\|^{2^*_s}}.
\end{aligned}$$

Now

$$\begin{aligned}
&\mathcal{I}_{\lambda,\mu}(t^-((\bar{w}_k, \bar{z}_k))(\bar{w}_k, \bar{z}_k)) = \frac{1}{2} a (t^-((\bar{w}_k, \bar{z}_k)))^2 (\|\bar{w}_k\|^2 + \|\bar{z}_k\|^2) \\
&+ \frac{1}{4} b (t^-(\bar{w}_k, \bar{z}_k))^4 (\|\bar{w}_k\|^4 + \|\bar{z}_k)\|^4) - \frac{(t^-((\bar{w}_k, \bar{z}_k)))^q}{q} \lambda \int_\Omega f(x) |\bar{w}_k(x,0)|^q\, dx \\
&- \frac{(t^-((\bar{w}_k, \bar{z}_k)))^q}{q} \mu \int_\Omega g(x) |\bar{z}_k(x,0)|^q\, dx \\
&- \frac{2(t^-((\bar{w}_k, \bar{z}_k)))^{2^*_s}}{2^*_s} \int_\Omega |\bar{w}_k(x,0)|^\alpha |\bar{z}_k(x,0)|^\beta dx \to -\infty \ \text{ as } \ k \to \infty,
\end{aligned}$$

this contradicts that $\mathcal{I}_{\lambda,\mu}$ is bounded below on $\mathcal{N}_{\lambda,\mu}$.

Let

$$l_0 = \frac{|c^2 - \|(w_0, z_0)\|^2|^{\frac{1}{2}}}{\|(w_{\epsilon,\eta}, w_{\epsilon,\eta})\|} + 1,$$

then

$$\begin{aligned}
&\|(w_0 + l_0 w_{\epsilon,\eta}, z_0 + l_0 w_{\epsilon,\eta})\|^2 = \|w_0\|^2 + \|z_0\|^2 + 2(l_0)^2\|w_{\epsilon,\eta}\|^2 \\
&+ 2l_0\langle w_0, w_{\epsilon,\eta}\rangle + 2l_0\langle z_0, w_{\epsilon,\eta}\rangle > \|(w_0, z_0)\|^2 + |c^2 - \|(w_0, z_0)\|^2| \\
&+ 2l_0\langle w_0, w_{\epsilon,\eta}\rangle + 2l_0\langle z_0, w_{\epsilon,\eta}\rangle > c^2 > t^- \left(\left(\frac{(w_0 + l_0 w_{\epsilon,\eta}, z_0 + l_0 w_{\epsilon,\eta})}{\|(w_0 + l_0 w_{\epsilon,\eta}, z_0 + l_0 w_{\epsilon,\eta})\|}\right)\right)^2,
\end{aligned}$$

that is, $(w_0 + l_0 w_{\epsilon,\eta}, z_0 + l_0 w_{\epsilon,\eta}) \in W_2$.

Proof of Theorem 1.1 (ii). Let us fix $\Gamma_{00} = \min\{\Gamma_0, \Gamma_3, \Gamma_*\}$ (we recall that Γ_3 is defined in Proposition 3.9, Γ_0 defined in the proof of Theorem 1.1 and Γ_* in Lemma 6.1) and define a path connecting W_1 and W_2 as $\gamma_0(t) = (w_0 + t\, l_0 w_{\epsilon,\eta}, z_0 + t l_0 w_{\epsilon,\eta})$ for $t \in [0,1]$. Then there exists $t_0 \in (0,1)$ such that $(w_0 + t_0\, l_0 w_{\epsilon,\eta}, z_0 + t_0 l_0 w_{\epsilon,\eta}) \in \mathcal{N}^-_{\lambda,\mu}$. Therefore, by Lemma 6.1,

$$\Theta^-_{\lambda,\mu} \le \mathcal{I}_{\lambda,\mu}(w_0 + t_0\, l_0 w_{\epsilon,\eta}, z_0 + t_0 l_0 w_{\epsilon,\eta}) < c_{\lambda,\mu}$$

for $0 < \Gamma < \Gamma_{00}$. Now from Proposition 3.9 (ii), there exists a Palais–Smale sequence $\{(w_k, z_k)\} \subset \mathcal{N}^-_{\lambda,\mu}$ such that $\mathcal{I}_{\lambda,\mu}((w_k, z_k)) \to \Theta^-_{\lambda,\mu}$. Since $\Theta^-_{\lambda,\mu} < c_{\lambda,\mu}$, by Proposition 4.1, upto a subsequence there exists (w^0, z^0) in $\mathcal{H}(\mathcal{C})$ such that $(w_k, z_k) \to (w^0, z^0)$ strongly in $\mathcal{H}(\mathcal{C})$. Now using Corollary 3.7, $(w^0, z^0) \in \mathcal{N}^-_{\lambda,\mu}$ and $\mathcal{I}_{\lambda,\mu}(w^0, z^0) = \Theta^-_{\lambda,\mu}$. Therefore (w^0, z^0) is also a solution. Moreover, $\mathcal{I}_{\lambda,\mu}(w, z) = \mathcal{I}_{\lambda,\mu}(|w|, |z|)$, we may assume that $w^0 \ge 0, z^0 \ge 0$. Again using the similar argument as in case of first solution and strong maximum principle (see Lemma 2.6 of [8]) we conclude that $w^0 > 0, z^0 > 0$ is a second positive solution of the problem $(S_{\lambda,\mu})$. Since $\mathcal{N}^+_{\lambda,\mu} \cap \mathcal{N}^-_{\lambda,\mu} = \emptyset$, (w_0, z_0) and (w^0, z^0) are distinct. This proves Theorem 1.1. □

Appendix

Proof of inequality (6.1). To prove (6.1), it is enough to show that

$$(1+m)^p \ge 1 + m^p + pm + C_1 m^{p-1} \tag{6.6}$$

for $m > 0$. Equivalently, it is enough (divide (6.6) by m^p and put $t = 1/m$) to prove the following inequality to show (6.6):

$$(1+t)^p \ge t^p + 1 + pt^{p-1} + C_1 t \tag{6.7}$$

for $t > 0$. Let us introduce $\mathcal{G} : (0, \infty) \to \mathbb{R}$ as

$$\mathcal{G}(t) = \frac{(1+t)^p - t^p - 1 - pt^{p-1}}{t}.$$

Then it is easy to show that $\mathcal{O}(t) = (1+t)^p - t^p - 1 - pt^{p-1} > 0$ for $t > 0$ and $p > 2$ (one can prove by showing the monotonicity of $\mathcal{O}(t)$ and observing $\mathcal{O}(0) = 0$). Now we have the following behavior of $\mathcal{G}(t)$:

$$\lim_{t\to 0^+} \mathcal{G}(t) = p \quad \text{and} \quad \lim_{t\to +\infty} \mathcal{G}(t) = +\infty. \tag{6.8}$$

From (6.8), using the continuity of $\mathcal{G}(t)$, there exists some $C_1 > 0$ (may depend on p) such that $\mathcal{G}(t) \geq C_1$ which proves (6.7) and consequently (6.1). $\square$

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